PHYSIOLOGICAL AND BIOMECHANICAL PERSPECTIVES ON ELBOW FUNCTION DURING FOREHAND AND BACKHAND DISC GOLF THROWS: A NOVEL INVESTIGATION

by

Lauren Rachelle Giles

Liberty University

A Dissertation Presented in Partial Fulfillment

Of the Requirements for the Degree

Doctor of Philosophy

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ABSTRACT

Over 81% of players have reported injuries linked to disc golfing, predominantly due to overuse. Since disc golf originated, there is a lack of research on the proper biomechanics of the disc golf throw. This descriptive, quantitative study seeks to be the first. We investigated 6 females and 35 males, aged 21-66, registered to compete in disc golf. The goal was to examine the kinetic parameters of healthy disc golfers to identify biomechanics likely associated with safety. Research questions investigated whether there is a connection between lean body mass and disc distance traveled, elbow torque and disc distance, or grip strength and torque on the elbow. After recruitment from tournaments, biometric data including handgrip strength (dynamometer), body mass index, and body fat percentage (bioelectrical impedance assay), were collected. Torque was recorded using a Motus sensor during forehand and backhand throws. Assumption tests revealed that lean body mass, maximum backhand and forehand distance and handgrip MVIC were normally distributed, while torque on either style of throw were not. A two-tailed Pearson correlation indicated that while some relationships were observed, none were statistically significant, leading to a failure to reject the null hypotheses. Factors such as individual variation and form might explain this, although notable findings emerged. Specifically, some torque measures surpassed the upper limits established by studies in other sports. There are numerous opportunities for future research to build on the findings of this study including introducing a longitudinal aspect, expanding to other sports with different throwing forms, and expanding to torque with “swinging” sports such as tennis or golf.

Keywords: disc golf, elbow torque, motus, elbow biomechanics, handgrip MVIC

Dedication

I want to dedicate this accomplishment to Laurie Ennis, my first professional mentor. I am so grateful that her levity, patience, lightheartedness, and impartiality imprinted on me before she left this Earth, and I miss her every day. Eternally humble, Laurie would roll in her grave to think I worked so hard for this to then dedicate it fully to her, so: this is also for Me. I want to dedicate this to the past version of Me who would have never predicted we'd be here (for so many reasons), and the future version of Me (who I can only hope will be using retirement to adopt dogs from hospice to give them the best final days on a tropical island).

Finally, this is for all of the powerhouse women who came before me. Your stories of wishing you did it, or did it sooner, instilled such a sense of urgency in me. I am so grateful to you as I stand here, completing a PhD. You all know who you are.

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List of Abbreviations

DEXA - dual energy X-ray absorptiometry (Tornero-Aguilera et al., 2022)

MVIC - maximal voluntary isometric contraction (Cronin et al., 2017)

LBM – lean body mass (Campa et al., 2024)

PAR-Q+ - physical activity readiness questionnaire for everyone (Warburton et al., 2011)

PDGA - professional disc golf association (Woods, 2019b)

UCL - ulnar collateral ligament (Solomito et al., 2015)

CHAPTER ONE: INTRODUCTION

Overview

Disc golf was founded in the 1970s and has developed to involve over two million athletes (Nelson et al., 2015). The average disc golfer is a middle aged, white, affluent man in the Midwest, but disc golf is beginning to reach more communities, and technology is making it accessible to people who cannot (now or ever) exercise safely in higher impact forms (Glattke et al., 2019). Part of this growth can be attributed to the reach of social media (Woods, 2019a) as well as the cost of "admission" being free to play in public parks (Woods et al., 2023), and the only equipment needed is a disc that can be as inexpensive as 5 USD (Nelson et al., 2015). However, many players experience injuries surrounding their disc golf participation, and there is no evidence on proper biomechanics of disc golf throwing for players to mimic to help reduce injury. Thus, healthcare providers treating injuries cannot advise for or against specific mechanics as part of an evidence-based practice. In addition, athletic trainers, personal trainers, strength and conditioning coaches do not have a working background knowledge to access in helping individuals develop strength and competencies in the right areas to avoid disc golf injuries. This study aims to investigate kinetic parameters of self-reported healthy disc golfers that have enrolled themselves in a disc golf tournament to gain information about common biomechanics that are likely to be safe, as the studied disc golfers employing them have no present or recent injuries.

Background

Disc golf was invented in the 1960s and has only continued to grow (Woods, 2019b). It is considered a sport that is widely accessible, because roughly 90% of disc golf courses are positioned in public parks, with the first official course being developed in Los Angeles' Oak

Grove Park in 1975. By 2018, up to 6,316 courses were constructed and the Professional Disc Golf Association (PDGA) was incepted and amassed nearly 37,000 members to participate in its over-3000 annual sanctioned tournaments. Most disc golfers are estimated to be recreational players who do not choose to join the PDGA, so it is assumed that significantly more people are disc golfing than is reflected in these statistics.

Evidence to the fact that disc golf is growing in number of players, it is also growing in the number of courses to support all of these disc golfers. Where in 1975 there was one course; the original, in Pasadena, California, between 2008-2019, the professional disc golf association (PDGA) reported an addition of 79 new courses per year for the following thirty years (Woods et al., 2023). By 2019, there were a total of 8,463 courses, and disc golf had gained popularity in several other countries including Finland, Sweden, and Japan (Woods et al., 2023) as well as in Bulgaria (Dimitrova, 2024) and Denmark (Rahbek and Neilsen, 2016).

Many studies have shown that when neighborhood poverty increases in urban areas, proximity to parks and green spaces decreases (Woods et al., 2023). The opposite trend is observed in rural areas. As such, when controlling for population size, disc golf course access is highest in the suburban Midwest and lowest in the urban areas of the east and west coasts. The trends show that U.S. disc golfers are found to be affluent, disproportionately male (roughly 94%), white (91%), younger (mean age of 33), and possess some college education at a rate lower than the national average (36%). This may be due to the social and communal aspects of disc golf, where social media is an established form of promotion participation, therefore making it more likely that people who are connected to each other and in geographically similar distributions to disc golf together (Woods, 2019a). Further evidence of the spirit of community

and volunteerism shows in the finding that when disc golf courses are designed, 65% of course designers and 74% of course installers are never paid for their work (Woods, 2019b).

The game of disc golf has few rules, so it is easy to learn and begin playing; as long as a person has a disc to use, it is otherwise cost-free to play (Woods, 2019b). It is also low-impact and is not highly physically demanding, which makes it accessible to more people who cannot exercise in other manners due to physical limitations. One such population is the visually impaired community. There are current initiatives to design enhanced disc golf courses to make it more plausible that individuals with visual impairments could participate (Glattke et al., 2019). Where other sports are not capable of inclusive participation, disc golf can be targeted for additions of audio and haptic feedback sensors that would make the course accessible. This could promote physical activity in the visually impaired and improve their health outcomes. Another population that benefits in a similar manner is the immunologically compromised. The covid-19 pandemic taught us that social distancing can protect vulnerable individuals from illness (Woods et al., 2023). Where it may not be safe for some people to exercise indoors in close quarters with others, these individuals may benefit from the outdoor, distanced setting promoted in disc golf.

While disc golf has been shown to have a positive impact on public health by encouraging people to become physically active, get outside, and form communities, it also has a positive impact on the environment. First, there is little need for deforestation to create courses as it is common for courses or holes to have their own unique obstacles (Rahbek and Nielsen, 2016) and disc golf takes place on comparably small plots of land when compared to traditional golf (Woods et al., 2023). There is also a reduced need for fertilizers and use of water to maintain the courses. Some studies have shown that disc golfers are open minded to the idea of playing in areas that make the sport more accessible in urban areas, such as abandoned factories.

Course design this way could reduce some environmental pressures associated with de-industrialization.

With all factors considered, disc golf is a way for individuals to become or stay active and therefore, is increasing in popularity to over two million players (Rahbek and Nielsen, 2016). As participation in physical activity increases, likelihood of injury also increases, but there is an insufficient number of studies to date that investigate the connection between disc golf biomechanics and type or frequency of injury. One study by Rahbek and Nielsen (2016) found that, of 105 surveyed Danish disc golf players, 39% reported injury during their career. The most commonly injured locations were the shoulder (31%) and the elbow (20%), and most injuries (93%) were due to overuse, or an onset that was gradual in nature. While the majority of injuries (92%) occurred on the throwing side of the body, all (8%) of the non-throwing side injuries occurred to the lower body. Perhaps more alarming, the median duration of symptoms in injuries current to the study was found to be nearly eight months, resulting in consequences including lost work and other impacts on life activities. All of these currently injured players were still playing disc golf for at least one hour per week over the course of the last month, suggesting that their condition may not have been improving due to continued aggravation of the injury.

A similar study was conducted by Nelson et al. (2015), although on a much larger scale. Their survey reached 883 disc golf players who answered 49 questions about their demographics, level of experience, and history of injury. These respondents reported some variety of injury over their career (81%), and while the most commonly injured locations matched those found by Rahbek and Neilsen (2016), the Nelson et al. study found that the elbow (46.0%) was most common followed by the shoulder (43.1%). These respondents reported that repetitive stresses

accounted for 57.4% of their injuries, where 29.4% were acute in nature (traumatically during a single throw, slip/loss of balance/fall), and 57.6% reported effects on the activities of their daily life, including work (34.5%). Regrettably, comparisons cannot be drawn to the Rahbek and Nielsen (2016) conclusions on the parameters of injury duration and affected side of the body as the Nelson et al. study did not evaluate either parameter. Other limitations include that the Nelson et al. (2015) study was posted for only 31 days and on the PDGA website. Although the authors state that this was during the World Championship tournament which increases website traffic, it is likely that a longer duration study would have returned greater outcomes. In addition, the demographics of respondents echoed the disc golf demographics in that 92.9% were male and 64% were between the ages of 25 and 44. While it was not explicitly stated, if the respondents' other demographics follow this trend, it is likely that they were also white, middle class or wealthier, located in suburban regions, and with about one-third having some college education (Woods et al., 2023). This does not give a good representation of injuries likely sustained by Americans, but rather, a very narrow subset of Americans. The study would have been stronger in terms of injury prevalence and future injury prediction if more women and racial minorities were included, but this would require the intentional selection of participants who do not represent the majority of disc golfers' demographics.

Problem Statement

Disc golf is a sport that has not been the scope of much scientific research. There are a handful of studies that place emphasis on investigating the sociological aspects of disc golf (Woods, 2019a) (Woods, 2019b) (Woods et al., 2023) (Glattke et al., 2019). None of these, however, evaluate parameters of body composition of disc golfers and outcomes in terms of sport performance. Anecdotally, some disc golfers believe in an association between grip

strength and throwing distance among disc golfers, which has not been evaluated in any formal scientific measures. A positive association has been found in previous studies between grip strength and sport performance (Cronin et al., 2017), although these studies do not specifically reference disc golfing. Cronin and colleagues (2017) found that there is a small, nonsignificant relationship (7%) observed between handgrip strength and throwing a projectile in sport, though the authors warn that it is necessary to test hand grip strength in a sport-specific manner in order to ascertain an accurate understanding of this relationship.

With respect to body composition of the athlete, it is believed that elite athletes' performance benefits from a reduction of body fat (Hector and Phillips, 2018). However, if body fat is intentionally reduced in this population, care should be taken to ensure that lean body mass is not decreased. With a reduction in skeletal muscle, it is believed that performance will suffer. However, there are over 2 million disc golfers (Rahbek and Nielsen, 2016), and a small proportion would likely be considered elite athletes. A comparison can be made between disc golfers and discus throwers (Liu and Yu, 2021), whose body composition has been evaluated. A study by Hirsch et al. (2016) on National Collegiate Athletic Association (NCAA) Division I track, and field athletes (mean age 19.2 ± 1.4 years) showed that throwers had a weight of 90 ± 18.3 kg, lean mass of 64.2 ± 11.7 kg and body fat of 23.6% ± 7.8%. This study was limited in that the sample size was 9, and women were not separated from men. When the athletes were evaluated simply by sex and not separated by event, the difference between women and men were found to be statistically significant in all of these parameters. However, the throwers had the highest weight, lean mass, and body fat percentage ($p \leq 0.05$) when compared against all other categories (sprinters, jumpers, pole vault/javelin, heptathletes/decathletes, and middle-distance runners/hurdlers). Finally, Hirsch and colleagues (2016) suggest that athletes with a

greater amount of lean mass may have improved performance and injury prevention. While this has not been tested yet in disc golfers, it lends some credibility to the hypothesis that athletes with larger bodies and with larger muscle mass may be well adapted to throwing. The problem is that with the combined factors of a lack of biomechanical research of disc golf and the fast growing nature of the sport, we cannot advise disc golfers on safe mechanics, train them for sport-specific movements in a manner that is based on literature, and know which mechanics, if any, are associated with improved performance.

Purpose Statement

The purpose of this descriptive, quantitative study was to evaluate experienced male and female disc golfers who self-identify by PAR-Q+ (Warburton et al., 2011) as uninjured, to gain information about their throwing biomechanics in an attempt to identify commonalities. Experienced disc golfers will be recruited for participation at disc golf tournaments that they enrolled themselves in. Body composition parameters of specific interest are lean mass percentage, determined by weighing the subjects and using bioelectrical impedance analysis (BIA) to identify lean body mass from fat mass (Duan et al., 2021), as well as hand grip strength assessed by dynamometer (McGrath et al., 2021). During three forehand and three backhand throws, disc golfers will wear a motus sleeve to measure varus and valgus torque at the elbow (Carroll and Hansen, 2019), and disc distance traveled will be measured using a laser rangefinder (Suh, 2019). Hence, the independent variables are the individual demographics of each disc golfer (body mass index (BMI), body fat percentage, lean mass, hand grip strength) as well as the type of throw (forehand versus backhand). Dependent variables include disc distance traveled and torque at the elbow.

Significance of the Study

The study's significance is that this research will be making the first meaningful contribution to the body of biomechanical evidence supporting the growing sport that is disc golf. This study is meaningful because disc golf is widely accessible to individuals of differing ages, abilities, and diagnoses (Glattke et al., 2019). In populations who cannot engage in physical activity as easily - for instance, the visually impaired, it is valuable to understand the risks of injury associated with more accessible forms of exercise such as disc golf. Individuals of varying income levels can also partake in disc golf, as participation costs are as low as 5 USD; the price of one disc (Nelson et al., 2015). As such, adding to the database for guidance on form and mechanics utilized by healthy disc golfers will help provide information for evidence-based practice of clinicians and trainers dealing with individuals who have sustained disc golf injuries.

In other sports such as baseball, it has been established that certain movements and variation within movement patterns make an individual more susceptible to injury (Miyazaki et al., 2021). For instance, changes in knee or elbow flexion upon release of a baseball can affect trunk rotation and other mechanics within the kinetic chain that increase torque on the shoulder and elbow joints and subsequently, injury at these locations. If such parameters exist for disc golf throwing, either for backhand or forehand style, it is not presently reflected in the literature. With that said, a Nelson et al. (2015) study found that 81% of surveyed disc golfers reported sustaining an injury and approximately one third of them did not receive any treatment for their injury. Where some level of injury can be expected in any setting due to accidents such as slips and falls, they only accounted for 9.4% of injuries. The major causative mechanisms were cumulative stresses (57.4%) followed by a single throw, step or movement (29.4%). More than half of injured disc golfers (57.6%) reported an impact on their daily life and more than one in

three (34.5%) reported an impact on their ability to work. Rahbek and Nielsen (2016) found that injured disc golfers experienced symptoms for a median of 240 days. If work and daily tasks are interrupted in a high proportion of cases (Nelson et al., 2015), it can be interpreted that these individuals also could not participate in exercise for a prolonged amount of time, which could have a negative impact on their overall physical health (Taylor et al., 2019), mental health (Smith and Merwin, 2021), or both.

As disc golf grows, it also grows more likely that diagnostic and rehabilitative facilities will begin to see individuals with related injuries in their practice. Therefore, it is necessary to develop a basis of knowledge regarding common throwing styles and biomechanical events that take place over the course of a normal throw so abnormal and risky deviations in form can be identified and corrected to avoid significant downtime.

Research Questions

RQ1: Is there a connection between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player?

RQ2: Does increased grip strength in experienced disc golf players translate to decreased amount of torque on the elbow joint during disc golf throws?

RQ3: Does the elbow joint of experienced disc golfers who throw a further distance experience higher amounts of torque?

Hypotheses

The null hypotheses for this study are:

H_01: There is no statistically significant relationship between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player.

$H_0$2: There is no statistically significant relationship between grip strength of experienced disc golf players and amount of torque experienced by the elbow joint.

$H_0$3: There is no statistically significant relationship between the amount of torque experienced by the elbow joint in experienced disc golf players and disc distance traveled during disc golf throws.

The alternative hypotheses for this study are:

H_A1: Disc golf players with higher amounts of lean body mass will have a further disc distance traveled compared to throws made by a player with a comparatively low amount of lean body mass.

H_A2: Players with higher grip strength will experience a higher amount of torque at the elbow joint than players with a comparatively lower grip strength.

H_A3: Players with a higher amount of torque experienced at the elbow joint will have a further disc distance traveled compared to throws made by a player with comparatively low amounts of torque at the elbow.

Definitions

Valgus rotation - lateral rotation; rotation away from the midline of the body. (Celli et al., 2021)

Varus rotation - medial rotation; rotation toward the midline of the body. (Celli et al., 2021)

CHAPTER TWO: LITERATURE REVIEW

Overview

Disc golf is a sport similar to golf in that its players throw discs at a target, similar to the way that golfers hit a ball toward a hole and strokes are counted (Giljarhus et al., 2022). As such, fewer throws to get the disc to the target is advantageous and therefore, it is worthwhile to throw the disc far, with accuracy. While there are some unusual throws used in disc golf, most fall into one of three categories. A Nelson et al. (2015) study found that most individuals use backhand (86.2%), with forehand being the second most common (12.7%) and overhead (1.1%) throws being relatively rare. As with any other physical activity, there is risk of injury. Having been formalized in the 1970s (Rahbek and Nielsen, 2016), disc golf is a newer sport. It benefits by increasing in popularity after the advent of social media and in 2017, had an estimated disc golfer population between 0.5 million (Woods, 2019a) and over 2 million (Rahbek and Nielsen, 2016). However, over 81% of disc golfers reported injury that could be attributed to disc golfing in a finding by Nelson et al. (2015). The most common diagnoses were a strained/pulled muscle (36.5%), sprain/ligament tear (24.5%), and tendinitis (22.0%). Of those, more than half reported that their injury had a negative impact on the daily activities of their life (57.6%) and many reported an effect on their work (34.5%). Finally, with the most common mechanisms for injury being overuse (57.4%) and a traumatic throw or movement (29.4%), rather than those that are accidental (for instance, a fall/slip/loss of balance (9.4%)) it is likely that a majority of future injuries can be avoided with preventative training and awareness of movements that are biomechanically friendly. As such, it is important to establish a scientific basis for common biomechanics during throwing in disc golfers that do not experience injury, the differences

between forehand and backhand throws, as well as using the findings and other literature to inform disc golfers on the safest ways to participate in the sport.

Conceptual Framework

As disc golf is an emerging sport, there is not much background information to help participants to understand the risks and contraindications for participation (Woods, 2019b). As such, analyzing biomechanics regularly employed in disc golf as they overlap with motions that are common to other more well-researched sports is a useful comparison. Certainly one of the most common contraindications in participating in any athletic activity is the risk for injury (Nelson et al., 2015). An epidemiological study of 874 disc golfers by Nelson et al. (2015) found that 81.8% had experienced an injury while disc golfing. The most common injuries incurred, in order of prevalence, included elbow injuries (46.0%), shoulder injuries (43.1%), back injuries (30.8%), knee injuries (28.1%) and ankle injuries (21.2%).

In order to understand what can increase the risk of injury, it is important to first consider the anatomical structures at each of the commonly injured areas.

Elbow mechanics and injuries

Anatomy of the elbow

The elbow is a complex joint that involves three bones that make up two articulations (Celli et al., 2021). The humerus is the single bone of the upper arm. The anterior component of its distal end features two condyles that each articulate with the bones of the forearm; laterally, the radius (forming the radiocapitellar joint) and medially, the ulna (ulnohumeral joint). This articular surface of the elbow is positioned at a slight (6°) natural valgus (lateral) rotation. The radius and ulna also articulate near the distal elbow, forming the proximal radioulnar joint. The

anatomical design of the elbow dictates the type of movements that can be performed; the simultaneous flexion and extension of the radiocapitellar joint and ulnohumeral joint flexes and extends the elbow joint in a hinging motion while medial and lateral rotation of the radius over the ulna at the proximal radioulnar joint provides supination and pronation, respectively. The strength of the ulnohumeral joint is demonstrated by features on the posterior surface, where the olecranon process of the ulna “latches” into the olecranon fossa of the distal posterior humerus. The inferior aspect of the olecranon fossa is formed by a feature called the trochlear notch, into which the olecranon is positioned. During flexion and extension, the olecranon rotates slightly within the trochlea in a motion that mimics turning a key in a lock, as the elbow is found to naturally be slightly valgus during extension and slightly varus during flexion (Celli et al., 2021). This allows the derivation of a carrying angle, referring to the angle found between the long axes of the humerus and ulna within the frontal plane. The carrying angle is a function of both musculoskeletal size and morphology of the radial head, with adult males averaging between 11-14° and adult females averaging 13-16°.

The bones that make up the elbow joint rely on ligamentous structures as well as surrounding muscles to provide strength through its two degrees of motion (Celli et al., 2021). The medial edge of the elbow joint is stabilized by the medial collateral ligament (MCL) whose anterior and posterior bundles run from the medial epicondyle of the distal humerus to the ulna in a v-shaped orientation that forms somewhat of a triangle when including its third aspect; the transverse ligament. While the transverse ligament is not currently believed to play a significant role in elbow stabilization, the anterior bundle of the MCL is the main stabilizer to valgus torque sustained by the elbow joint. The elbow is secondarily stabilized against valgus torque by the posterior bundle of the MCL, although its main role is to prevent hyperflexion.

The lateral collateral ligament (LCL) arises from the lateral epicondyle of the humerus and spans the lateral skeletal structures of the elbow joint (Celli et al., 2021). Like the MCL, it is also composed of three aspects; the radial collateral ligament and lateral ulnar collateral ligament (UCL) which terminate on their respective bones, and the annular ligament. The annular ligament forms on the radial part of the LCL, arcs over the radius, and terminates on the ulnar part, securing the proximal radioulnar joint. As the radial head is not perfectly round, the annular ligament tightens on either the anterior or posterior end during movement to support supination and pronation, respectively. Simultaneously, and with help from the UCL, the annular ligament helps resist subluxation of the proximal radioulnar joint.

Elbow Movements

The movements of the elbow are also supported by four main groupings of skeletal muscles (Celli et al., 2021). Posteriorly, only the tendon of the triceps muscle crosses the elbow joint. Laterally, the elbow is stabilized by muscles that contribute to supination at the elbow and extension at the wrist and hand. This includes the brachioradialis, extensor carpi radialis longus, extensor radialis brevis, extensor carpi ulnaris and the anconeus, which also works with the triceps brachii to extend the elbow. The largest proportion of muscles crossing the elbow joint help to perform flexion of the elbow, or flexion at the forearm and hand. The prime flexors of the elbow are the brachialis, biceps brachialis, and brachioradialis which all attach on the anterior surface of the forearm surrounding the joint. Also, the anterior surface includes the pronator teres, flexor carpi radialis, palmaris longus, flexor carpi ulnaris, and flexor digitorum, which contribute to pronation at the elbow joint. Finally, the supinator conducts supination, with assistance from biceps brachii.

Mechanisms of Elbow Injury - Flexion & Extension

The importance of the UCL in resisting valgus forces during throwing biomechanics makes it one of the most commonly injured structures of the elbow in throwing athletes (Li et al., 2019). The UCL can be associated with elbow extension as the elbow is found to naturally be positioned slightly valgus during extension (Celli et al., 2021). During baseball throwing, this action would take place during the ball release phase of the throw (Camp et al., 2017) (Tanaka et al., 2020).

Mechanisms of Elbow Injury - Pronation & Supination

Normal range of motion for pronation and supination is roughly 80°, although it is important to compare to the contralateral side of the body as individuality plays a role in ligamentous laxity, among other components dictating movement (Laratta et al., 2017). It is more common that injuries would occur by excessive flexion and extension movements as compared to pronation and supination. However, repetitive or excessively stressful pronation and supination, as seen in swimmers or tennis players, can promote compression of the posterior interosseous nerve. This pathology presents as pain and/or weakness in the forearm, especially in the dorsal compartments of the forearm. Other sources of injury surrounding pronation and supination, though not necessarily solely attributable to either motion, can include radiocapitellar joint injury, osteochondritis dissecans, or lateral elbow plica.

Repetitive forearm pronation can result in a condition called golfer’s elbow, formally known as medial epicondylitis (Laratta et al., 2017). This occurs in overhead throwing athletes and is associated with valgus loading. However, lateral epicondylitis, or ‘tennis elbow’, is more common than medial epicondylitis. Both manifest with pain and decreased grip strength, but

neither typically occur strictly due to pronation or supination; they usually also require concurrent repetitive movements at the wrist.

Common elbow injuries in overhead athletics

Men's baseball predominantly features overhand throwing and experienced the most (n = 20) UCL injuries that were deemed non-contact (73.1%) in nature in a Li et al. (2015) longitudinal study. This equates to a rate of 1.12 per 10,000 UCL injuries requiring medical treatment. The majority of the injuries (n =14) did not require surgery and were not severe (n = 13) when severity was defined by the number of days excluded from participation. Of the injuries requiring surgery (n = 4), however, the majority were throwing injuries in baseball players (n = 3). Commonly, sprains and partial tears are incurred but do not require surgery. Surgical intervention is typically reserved for complete tears, which are less common in overuse injury as it is likely that an athlete will self-limit due to pain or poor performance before an overuse injury advances to a complete tear.

Elbow injuries of Disc Golfers

In an epidemiological survey study, Nelson et al. (2015) found the elbow to be the most injured body part, present in 46.0% of the 874 respondents. Of those, only 10.3% reported previous injuries to the elbow which indicates likelihood that disc golf is correlated to elbow injuries. The elbow is the only joint found to have a statistically significant increased rate of injury due to throwing style, where forehand throwers sustained elbow injury at an increased rate when compared to backhand throwers (p=0.014). Of the 112 individuals utilizing a forehand throw, 53 reported elbow injuries compared to 269 among the backhand throwers (761). Although it was the most common injured location, it was not reported that any of the 115 disc golfers who reported surgical treatment underwent elbow surgery, specifically.

Shoulder mechanics and injuries

Anatomy of the Shoulder

Four articulations comprise the shoulder girdle (Kadi et al., 2017). The glenohumeral, acromioclavicular (AC), sternoclavicular (SC), and scapulothoracic joints are formed between features of the humerus, scapula, clavicle, and sternum. The scapula is common to three of these joints, as it provides important features to each articulation. Laterally, it provides the glenoid fossa, into which the proximal humerus settles to form the glenohumeral joint. Posteriorly, the inferior angle of the scapula adheres to muscles including the serratus anterior, inferior component of the trapezius, latissimus dorsi, and the rhomboids (Di Giacomo et al., 2023). As such, it is not an osseous articulation like the other three joints of the shoulder. The acromion is a postero-lateral scapular process which articulates with the lateral end of the clavicle to form the AC joint (Kadi et al., 2017). The medial end of the clavicle articulates with the sternum to form the SC joint. Together, these joints contribute to the movements of the shoulder which allow it to be the most mobile joint in the human body.

To strengthen the joint, the shoulder's anatomy contains heavy reinforcement by soft tissues. Inside the glenoid fossa is the glenoid labrum, which runs the circumference of the fossa to add surface area without sacrificing range of motion (Kadi et al., 2017). It is also supported by the rotator cuff, whose four muscles and their tendons run from the humeral head to the anterior (subscapularis), supero-posterior (supraspinatus), infero-posterior (infraspinatus and teres minor) aspects of the scapula. Although it is not part of the rotator cuff, the shoulder joint is also stabilized, in part, by pectoralis major (Morell and Thyagarajan, 2016) as well as latissimus dorsi and the deltoids.

The osseous supporting joints to the glenohumeral joint, the SC and AC joints, are both found on the anterior surface of the body. The AC is a gliding joint surrounded by the acromioclavicular ligaments, coracoacromial ligament, and coracoclavicular ligaments (Kadi et al., 2017). The SC joint is the least commonly dislocated joint of the shoulder at a frequency of only 3% (Morell and Thyagarajan, 2016). The reason for this is due to its structure; it is a diarthrodial joint allowing elevation/depression and protraction/retraction of the shoulder girdle and forming the only bony articulation where the upper limb attaches to the axial skeleton. As the articulation would be unstable on its own, it is reinforced by soft tissue such as the rhomboid ligament which functions to resist superior and inferior rotation. The subclavius runs from the first rib to the inferior clavicle, supporting it against excessive upward movement. Finally, the interclavicular ligament, the posterior and anterior sternoclavicular and capsular ligaments provide strength around the joint's surface.

Biomechanics of throwing a disc at the shoulder

With respect to the shoulder, a study by Koeble and Seiberl (2017) found that ultimate frisbee throwers experienced a statistically significant difference in range of motion between internal rotation on their dominant arm but not on their external rotation. They also found a significantly higher maximum strength during external rotation that is unusual as previous literature had found other sports (baseball, volleyball, tennis) to experience a decrease or no change in external rotation. These translated to a glenohumeral internal rotation deficit and external rotation that Koebel and Sieberl then correlated to peak shoulder torque ($r = -0.69$, $p<0.01$). This fell within the previously reported accepted range for joint health and injury prevention; -0.66 to -0.75. These differences were not seen in the non-dominant arm, suggesting that these were sport specific functional and adaptations of the body to increase mobility.

Shoulder movements

The ball-and-socket joint of the shoulder, the glenohumeral joint, is considered the main joint of the shoulder (Kadi et al., 2017). It provides four general movements; flexion-extension, abduction-adduction, internal rotation-external rotation, and circumduction. As such, it is the most commonly dislocated joint in the body due to a trade-off that is commonly seen in anatomy; as degrees of movement increase, stability decreases.

Mechanisms of Shoulder Injury - External & Internal Rotation

In a meta study of overhead athletics, it was found that the shoulder movements tend to increase external rotation and decrease internal rotation of the shoulder joint due to the velocity and range of motion experienced at the joint (Tooth et al., 2020). Changes to the shoulder have been experienced over as short a time as a few months; one athletic season. Another general issue is scapular dyskinesis; the incorrect movement or resting position of the scapula. This could be attributed to either a decrease in the ratio of external rotators to internal rotators, a result of sport-specific strength modification in the muscles surrounding the shoulder joint due to increased use when practice begins. For other overhead sports, it was found that an internal rotation deficit up to 20° was permissible but deficit beyond that could be attributed to stiffness or an increased risk of injury. A difference in internal rotation range of motion between the throwing and non-throwing arm greater than 13° was also identified as a risk factor for increasing injury by up to six times the risk when side-to-side difference was less than 13°. With respect to external rotation, a side-to-side difference of just 5° or greater or less than 93° at the throwing shoulder was associated with an increased risk of up to four times above baseline. Extrapolating this to disc golf throwing, these findings suggest that motions that induce

excessive muscular imbalance and alterations in scapular range of motion could be contributing to shoulder injuries in disc golfers.

Mechanisms of shoulder injury - abduction & adduction

Horizontal adduction plays more of a role than abduction in injuries of overhead athletes (Cools et al., 2015). Deficits in range of motion tend to manifest in the form of posterior joint capsule stiffness. This is potentially the most common sport-specific adaptation seen in overhead athletes and is evidenced clinically by decreased range of cross-body adduction at the shoulder and a decrease in internal rotation. As such, internal rotation and adduction tend to be closely associated in the way their symptoms manifest. Posterior shoulder capsule stiffness is hypothesized to result from soft tissue microtrauma sustained during the deceleration phase of the throwing motion of the various overhead sports. This may, in turn, lead to shoulder impingement or contracture of the joint capsule and subsequent injury to the glenoid labrum (Cools et al., 2015).

Common shoulder injuries in overhead athletics

In overhead athletes, the most commonly reported shoulder injury is a strain, typically due to chronic overloading as opposed to a single traumatic event (Cools et al., 2015). Sport-specific adaptations and posture are other factors that affect the position of the shoulder joint and therefore, the kinetic chain. This pattern can lead to the development of the next most common shoulder injuries: glenohumeral internal-rotation deficit (GIRD), rotator cuff strength imbalance, scapular dyskinesis.

Shoulder injuries of Disc Golfers

Shoulder injuries accounted for 43.1% of the reported injuries in disc golfers in a Nelson et al., (2015) study; the second most common. Following their injury, 115 disc golfers required

surgery and of those, thirteen included rotator cuff repair or other surgery at the shoulder. In a similar epidemiological study of 105 disc golfers, Rahbek and Nielsen (2016) found that the shoulder was the most commonly injured joint (31%). Of the 17 athletes reporting shoulder injuries, 12 were gradual in onset, 5 required medical attention, and all were experienced on the throwing side. Based on previous studies (Kibler et al., 2013) (Lintner et al., 2008) (Sciascia et al., 2012) in other sports that show that the kinetic chain forms maximum force in the legs and trunk and transfers that energy to the shoulder, elbow and hand. Rahbek and Nielsen propose that if similar mechanisms are utilized by disc golf throwers, it can be hypothesized that a suboptimal kinetic chain could contribute to the high incidence of injured throwing-side shoulders (31%) of disc golfers.

Back mechanics and injuries

Anatomy of the Spine

The spine houses a number of structures aside from strictly the vertebral column (Frost et al., 2019). Within the vertebra is the spinal cord and the nerves that exit from it, but the spine also contains neural connections to muscles and ligaments that provide stability and control of the body. Finally, between each pair of vertebrae except the atlas and axis (C1 and C2) resides an intervertebral disc which provides cushioning, shock absorption, and weight distribution. These discs are sometimes compared to a jelly donut as they are composed of a tough exterior called the annulus fibrosus which is composed of fibrocartilage, and then a gelatinous center called the nucleus pulposus. Taken together, these discs can comprise up to 30% of the length of the spine, and their main component is water. They are innervated by branches of the spinal nerves, which

tend to move deeper into damaged discs over the course of time. As a result, any amount of pressure above normal load can contribute to back pain.

Back Movements

The spine is subdivided into five sections; the cervical spine (neck), thoracic spine (ribcage), lumbar spine (lower back), sacrum (which provides attachment for lower extremities), and the coccyx (commonly referred to as the "tailbone") (Frost et al., 2019). The cervical spine contains seven vertebrae numbered C1-C7. Along with their intervertebral discs, they perform flexion, extension, rotation and lateral flexion. The thoracic spine is composed of twelve discs and while it can still flex, extend, and rotate, it has reduced mobility in comparison to the cervical spine. The lumbar spine contains five vertebrae. Like the rest of the spine, they experience flexion, extension, rotation and lateral flexion. Unlike the rest of the spine and due to their inferior position, the lumbar vertebrae support more of the weight of the body and withstand a high amount of stress, strain, and torsion. For this reason, the five intervertebral discs at this location are naturally thicker and have the highest surface area of any of the discs, to support the lumbar spine through flexion, extension, and rotation.

Common back injuries in overhead athletics

Despite their compensatory structural design, the lumbar intervertebral discs are the most commonly injured as it has been documented to sustain up to thousands of Newtons in force (Frost et al., 2019). Under high amounts of force and mechanical strain, the annulus fibrosus can sustain small tears. These are typically in the posterior disc and can cause the nucleus to bulge in such a way that it presses on the spinal cord, spinal nerves, or both. This type of injury is termed a "bulging disc", unless the tear spans the full thickness of the annulus fibrosus which is then

termed a “herniated disc”. The second most commonly injured location is the intervertebral disc between C5/C6, which experiences these movements to the maximum degree.

Back injuries experienced by overhead athletes results mostly from inadequate strength balance in the core, specifically the lumbar core (Cools et al., 2015). As such, thoracic spine stiffness, hyperkyphosis, and hip range of motion and strength deficits can occur. It is believed that these issues contribute to a “cascade to injury” along with scapular dyskinesis, GIRD, and rotator cuff strength imbalance.

Back injuries in disc golfers

Back injuries were cited by 218 disc golfers in the Nelson et al. (2015) epidemiological survey-based study and while the study did not specify all causes of strains and sprains, spinal disc/nerve injuries were reported in 6.7% of respondents. It was not supported by the data that one style of throwing caused more back injuries than the other ($p = 0.209$) among the 22 forehand and 191 backhand throwers reporting back injury. When considering the neck as well (although there is more to the neck than just the cervical spine), 67 injuries occurred in backhand throwers which was not found to have statistical significance when compared to the number of forehand throwers with neck injuries (9, $p=0.788$). Further, of those reporting back injuries, seven required surgery that they attributed to their disc golf injury. However, of those reporting prior injury (64.8%), the back was the most common location for previous injury (20.2%) so it is possible that some were reinjured rather than sustaining a new back injury. This could also increase the probability of an individual requiring surgery.

Knee mechanics and injuries

Anatomy of the Knee

The knee joint is technically composed of three bones; the femur, tibia, and patella, despite the fibula being an anchoring point for some of the ligaments that support the joint (Hassebrock et al., 2020). Within the joint, there are two cruciate ligaments which cross between the condyles of the femur and tibia; the anterior cruciate (ACL) and the posterior cruciate ligaments (PCL). The ACL spans the joint cavity from the lateral condyle of the femur to the medial condyle of the tibia. Conversely, the PCL lies behind the ACL and runs the opposite direction; from the medial condyle of the femur to the lateral condyle of the tibia. These ligaments are responsible for stabilizing the knee during extension and flexion, respectively and also help to maintain the position of the tibia and therefore, the inferior portion of the knee joint. The ACL is tasked with supporting repetitive loads; its tensile strength has been documented to be up to 2200 Newtons, while the maximum tensile strength of the PCL is only 112.3 Newtons. The main role of the PCL, then, is stabilizing against excessive varus and valgus rotation of the joint. The ACL and PCL are supported by nearby muscles such as the popliteus, hamstrings muscles, and gastrocnemius (posteriorly) and quadriceps group and their tendon (anteriorly), as they are known to sustain up to as much as 24 times the individual's body weight during weight bearing activities such as running. In addition, the medial and lateral menisci are pads of cartilage that are positioned between the medial and lateral condyles of the femur and tibia that help to decrease stress at the joint by providing cushioning.

The knee joint also has a pair of collateral ligaments which span the medial and lateral edges of the joint named the medial collateral and lateral collateral ligaments (Hassebrock et al., 2020). The LCL runs from the lateral edge of the femur's lateral epicondyle to the head of the

fibula, while the MCL spans from the medial epicondyle of the femur to the medial condyle of the tibia. As such, they can also be called the fibular collateral and tibial collateral ligaments. Together, they support their respective menisci and the action of the PCL as the LCL functions to stabilize against excessive varus rotation of the knee and the MCL against excessive valgus rotation. The MCL also takes on some of the function of stabilizing the knee joint against excessive anterior movements and internal rotation. While the LCL receives support from the iliotibial band and the biceps femoris muscle, the MCL is reinforced anteriorly by the sartorius and posteriorly by the semitendinosus and semimembranosus.

Knee Movements

The knee can only perform flexion and extension, although there is a slight varus and valgus pivoting that is acceptable (Hassebrock et al., 2020). This pivoting motion is referred to as a screw-home mechanism (Albishi et al., 2024).

Knee injuries to the collateral ligaments

When injuries to the ligaments of the knee are observed, the most frequently affected (7.9% of cases) is the MCL (Hassebrock et al., 2020). These injuries are graded on the Hughston method (grade I-III) based on the amount of tenderness the patient experiences upon palpation and the degree of valgus rotation that can be performed. Fortunately, even with the most acute degree of MCL tears (grade III), surgical intervention is rarely necessary as long as movement is not hindered significantly. When surgery is required, however, good long term results have been reported. The mechanism of injury is typically when valgus stress is increased, as this is a role of the MCL and would hyperextend the ligament. This typically occurs in actions where a person suddenly changes direction or otherwise experiences a knee trauma.

Isolated injuries to the LCL are thought to be rare, although they are known to be misdiagnosed so they may be more frequent than presently believed. When they do occur, causative varus, rotational, extension, or translational forces are to blame regardless of if there was an impact sustained during the injury.

Similar to MCL injuries, LCL injuries are also graded but the grading system relies on magnetic resonance imaging (MRI) to assess the degree of tear from minimal (grade I) to complete (grade III). While non surgical treatments are recommended for isolated LCL tears regardless of grade, it is more often the case that LCL tears accompany others. In this case, due to poorer outcomes, grade III LCL tears are recommended for surgical repair within multiligamentous injuries when other ligaments also require surgery.

Knee injuries to the cruciate ligaments

Anterior cruciate ligament (ACL) injuries tend to be non-contact to low-contact injuries most (70%) of the time (Boden and Sheehan, 2021). These injuries tend to be acute opposed to cumulative, and the main causative agent seems to be a compressive axial force. Such force is achieved through the concerted effects of a strong, eccentric quadriceps contraction with concurrent knee abduction. This is likely due to the finding by magnetic resonance imaging (MRI) that landing from a jump or deceleration movement with a flat foot generates a higher tibial slope. A higher slope on the tibia allows the lateral condyle of the femur to slide forward within the joint, thus injuring or rupturing the ACL. These mechanistic findings are supported by the finding that it is reproducible in cadaveric studies.

Unlike with the MCL, ACL reconstruction surgery has not been reported to have good long-term results (Hassebrock et al., 2020). When injuries to either cruciate ligament occur, it is often in conjunction with a collateral ligament as they have common functionality.

Like with LCL tears, posterior cruciate ligament (PCL) injuries tend to occur concurrently with other ligamentous injuries (Bedi et al., 2016). The main function of the PCL is to limit the degree to which the tibia moves during knee flexion and its secondary functions are to decrease external, internal, and varus-valgus rotation. As such, common mechanisms of injury involve movements when these movements are exaggerated. These can be as simple as abnormal knee kinematics while climbing stairs or running for lesser degree injuries, while other injuries have occurred when a person incurs a force directed at the proximal tibia, especially during knee flexion. For instance, sudden hyperflexion, hyperextension, or falling onto a flexed knee can injure the PCL, especially when the foot is plantarflexed.

Knee injuries of disc golfers

Nelson et al. (2015) found that the most common surgery reported following disc golf injuries included knee surgeries including those for menisci (13 of 115) and knee scope (5 of 115). Individuals who utilized a backhand throw reported the highest amount of knee injuries (168 versus 28 with forehand), but it was not found that backhand throws cause more injuries than forehand ($p = 0.489$). In addition, 19.9% of respondents reported previous injury, making it possible that some aggravated a pre-existing injury rather than provoking a new one.

Ankle mechanics and injuries

Anatomy of the Ankle

The anatomy of the ankle joint contributes to its propensity for injury as it is comprised of only three bones (the distal tibia, distal fibula, and superior aspect of the talus - one of the tarsal bones) which are held together by a number of ligaments (Golanó et al., 2014). The ligaments involved either connect the tibia to the fibula (anteroinferior tibiofibular,

posteroinferior tibiofibular, and the interosseous tibiofibular ligament), the fibula to the tarsal bones (anterior talofibular, calcaneofibular, and the posterior talofibular ligament), or the tibia to the tarsal bones (medial collateral ligament, also called the deltoid ligament).

Ankle Movements

Despite its anatomical stabilization, the ankle is a hinge joint with one axis of normal movement; dorsiflexion and plantar flexion (Golanó et al., 2014). The ankle is not designed to sustain inversion and eversion. Subtalar eversion and inversion are the movements most commonly experienced in ankle sprains.

Ankle Injuries - plantar flexion

It is believed that up to 90% of ankle sprains are inversion sprains to the lateral ligaments of the ankle while the foot is plantar-flexed, most often the anterior talofibular ligament (ATFL) (Golanó et al., 2014). Hence, the medial collateral ligament is believed to only account for 5-15% of ankle injuries.

Ankle injuries of disc golfers

Ankle sprains are the most common ankle injury in athletics, as the ankle is one of the most commonly injured joints overall (Golanó et al., 2014). As disc golf is not a contact sport and Nelson et al. (2015) did not collect data on the mechanism of injury, it can be assumed that ankle injuries are not due to a slip and fall, which were only reported by 9.4% of respondents. Taken with the information provided by Golanó et al. (2014), ankle injuries likely occurred in a traumatic (during a single throw or step) event, during either the takeoff or release phases of a disc golf throw.

Among disc golfers, 21.2% reported ankle injuries in the Nelson et al. (2015) study, the fifth-most commonly injured location. Of those, 17.9% reported previous ankle injury and there

was not a statistically significant difference observed between injured forehand (125) or backhand (24) throwers ($p = 0.189$). However, of the 115 respondents that had undergone surgery, 7 reported foot or ankle surgery making the ankle the second most common to have been operated on.

Biomechanics of Disc Golf Throws

Ultimate frisbee is a flying disc sport with some established publications, which warrants comparisons between disc golfers even though the intensity of ultimate frisbee is more comparable to soccer while disc golf is more accurately compared to that of golf (Pulido and Lystad, 2020). A study by Winograd and Engsberg (2012) evaluated spin, velocity and accuracy with respect to different grips used in flying disc sports. While none of the throws showed a statistically significant difference in accuracy, the backhand throws were found to have more spin and wobble when compared to the forehand throw. The authors explain that this is advantageous to ultimate frisbee but this may be disadvantageous for disc golf, where players are trying to land the disc in a stationary goal basket in as few throws as possible (Glattke et al., 2018). This establishes a setting where players are trying to throw the disc as hard as possible without sacrificing accuracy, so as to have fewer “strokes” per hole and a better score. It is perceivable that throwing form which results in high distance would also promote high velocity.

Winograd and Engsberg (2012) found that there was no difference in linear velocity between forehand and backhand throws but that backhand throws had higher angular velocity. This is advantageous as discs thrown with high angular velocity have been shown in previous research to be more stable. Further, when grips between forehand throws were compared, the closed grip showed a higher linear velocity than split grip. While the techniques for these grips are not well established in peer reviewed literature, they are discussed in a Holcomb (2023)

thesis. The author states that during the backhand throw, the thumb is placed on top of the disc with the fingers curling around the end of the disc to grip it, called the “power grip”. It is common to extend a finger, like the index finger, out onto the edge to make changes to the amount of control that the thrower has over the disc (“single finger grip”). Another variation is what Holcomb calls the “split-fingers” grip, which involves the index finger being placed on the outer rim of the disc and the remaining fingers spread on the underside of the disc. The cited advantage to this is improved power. This is likely what Winograd and Engberg call the “split grip” on a forehand throw. As forehand throws are known to produce less force, it is plausible that players would make adjustments to try to increase flight distance. Therefore, the Winograd and Engsberg (2012) “closed grip” is likely similar to Holcomb’s (2023) “power grip” or “single-finger grip”.

In the absence of disc golf specific studies, disc golf throwing can also be compared to throwing discus within track and field. While a rotation is employed in discus throwing all the other components aside from grip (maximal back swing, right foot touchdown, left foot touchdown, release) overlap in form with disc golf throwing (figure A, Liu and Yu, 2021). There is an established history of discus data collection. In 1999, the Olympic gold medalist in discus had a recorded disc exit velocity of 30.80 m/s, while the silver medalist was 27.18 m/s (El Monsef et al., 2012). With a seemingly large difference between these elite competitors, it places importance on understanding the difference in form between throwers. A study by El Monsef et al. (2012) found that, in a comparison between three professional discus throwers, a main difference was in the release angle and the proportion of time spent during each phase of the throw. Top discus throwers have been documented to spend roughly 17% of their total throw in

the delivery phase. This is the portion of time where the disc is being released, so it can feasibly be compared to disc golf backhand throws.

Related Literature

Varus and Valgus Torque at the Elbow

As previously stated, the natural orientation of the skeletal features of the elbow joint hold it in a slight valgus position at extension and slight varus position at flexion (Celli et al., 2021). It is worth understanding what contributes to torque at the elbow as it is the most commonly injured joint in disc golfers (Nelson et al., 2015). To date, however, there has not been a study specifically evaluating torque on the elbow during disc golfing. However, one realm within which varus torque has been thoroughly investigated is within baseball; specifically, baseball pitching.

It was found in a study by Tanaka et al. (2020) that varus torque of the elbow was impacted by the actions of the other joints during the pitch. Across the pitchers evaluated, the mean peak elbow varus torque was 57.0 ± 15.7 Newton meters (Nm). There was a positive association between elbow varus torque and ball speed ($R^2 = 0.05$; $P < .001$) with the mean ball speed being (74.7 ± 5.5 mph). Tanaka et al. (2020) identified the five most impactful kinematic variables that contributed to a decrease in torque on the elbow. They included: increased wrist extension at the time of stride foot contact with the ground, upward displacement in the change in center of mass in a vertical plane during the stride phase, increased knee flexion on the leading leg at stride foot contact, increased knee extension on the trailing leg at stride foot contact, and elbow pronation at stride foot contact. This is in alignment with previous research (Werner et al., 2002) (Aguinaldo and Chambers, 2009) (Sabick et al., 2004) in baseball pitchers that found peak

elbow varus torque was most impacted by shoulder abduction at stride foot contact, peak shoulder horizontal adduction angular velocity, elbow flexion at the instant of peak elbow varus torque, and peak shoulder external rotation torque.

Also, within baseball pitching valgus (lateral rotation) torque has been assessed in detail. A study by van Trigt et al. (2023) investigated valgus torque at the elbow with emphasis on the humeroulnar joint gap and the UCL because adult (professional) pitchers are disabled due to elbow injury in 28% of cases. Due to the forces experienced by the body during pitching, this is almost an expected finding. The medial elbow is the location of the UCL, whose responsibility is, in part, to resist over-rotation in the lateral direction. During the course of a pitch, especially when the shoulder is at maximum external rotation, the elbow experiences great amounts of valgus torque; up to 50 Nm in healthy athletes.

One mechanism that has been suggested to protect the UCL during times of valgus torque is the concentric flexion of muscles dedicated to the flexion and pronation of the forearm (van Trigt et al., 2023). It is proposed that these muscles close the gap in the humeroulnar joint, thereby stabilizing the UCL. This is proposed due to the observed phenomena of the increase in the humeroulnar joint gap after youth pitchers threw 60 pitches, increasing even further after 100 pitches coupled with the finding that individuals with UCL injuries experience increased humeroulnar joint gap during valgus rotation, even at rest. During a pitch, the pronation-flexion muscles have been shown to contract at least 20% of their maximum voluntary contraction and up to 40%. It was found that applying handgrip force at 20% of one’s maximum was correlated to a slight decrease in humeroulnar joint gap and valgus stress on the elbow, but the difference was not statistically significant. This lack of statistical significance may be explained by the

findings in other studies (Tsubono et al., 2022) that have suggested a 50% maximum handgrip force is required to decrease the humeroulnar joint to a meaningful degree.

Arm Slot

Arm slot is a measure that is used in sports involving thrown projectiles. It is an angle that is derived from a combination of biomechanical factors including the degree of trunk tilt, abduction of the shoulder and flexion of the elbow (Manzi et al., 2023). Manzi et al. collected this information by placing a variety of motus sensors and 42 reflective markers on athletes' anatomical landmarks and had them throw a mean 11 pitches under an 8-camera system. The data were aggregated per pitcher and median values were derived to reduce noise or bias introduced by the athlete's total pitch count. An ANOVA was applied with a post-hoc Tukey test. When compared to pitching mechanics, overhand deliveries have the lowest degree of arm slot (37.1° ± 7.4°) while sidearm deliveries (63.2° ± 6.6°) have the greatest degree with three-quarter deliveries (49.7° ± 2.9°) falling in between. It appears as though these ranges are inconsistent among the literature as they overlap the distinctions made by Escamilla et al., (2018) that overhand ≤40° arm slot angle, three-quarter has a 41° to 69° arm slot, and the arm slot for sidearm is ≥70°.

It has been found relatively consistently among the literature that elbow flexion torque is higher in the three-quarter and sidearm groups than the overhand group, but that shoulder anterior force is higher in the three-quarter group compared to the sidearm (Manzi et al., 2023). High elbow flexion torque is disadvantageous as it has been associated with superior labrum anterior-posterior (SLAP) lesions at the shoulder joint and increased varus torque at the elbow. In addition, it was found that arm slot could be correlated to the timing of the maximum angular velocity obtained in the upper trunk. It is therefore perceivable that the increased arm slot

observed in overhand style can impact the kinetic chain in such a way that timing of pelvis-trunk separation is impacted, negatively affecting pitching form and ball exit velocity while increasing the risk of arm injuries. Conversely, sidearm throwing was correlated to an increased degree of elbow flexion in high school pitchers while being associated with decreased elbow varus torque and shoulder internal rotational torque in professional pitchers. This conflicts with previous findings that pitchers with an overhand delivery experienced significantly less elbow varus torque (46 ± 29 N·m) compared with pitchers who threw sidearm delivery (66 ± 24 Nm)(Camp et al., 2017). Hence, it was suggested by Manzi et al. (2023) that the most ideal arm slot may be 60°-70° but more research should be performed before sidearm is accepted as the least dangerous form.

Injuries, Overhead Baseball Throwing

When overhead throwing is considered, one well-researched comparison that can be drawn is to baseball, as all position players employ overhead throwing mechanics in baseball (Scarborough et al., 2020). The phases of throwing a pitch can be broken down into three components. First, the ‘cocking’ phase is where the ball is drawn behind the body, the elbow is flexed while the shoulder is horizontally abducted and externally rotated, and the trunk rotates contralaterally while extending slightly while the pelvis rotates laterally, “opening” the hips as the front leg is used to stride forward with the hip and knee flexed while the back leg remains planted (Camp et al., 2017)(Figure B, Tanaka et al., 2020). Next, during the acceleration phase, the elbow lags behind the trunk as the upper torso and pelvis begin to rotate medially and flex forward while the pitcher presses off from the ground with their back leg, simultaneously increasing hip flexion in the stride leg. At this time, peak varus torque can be experienced due to the high degree of shoulder external rotation, while the elbow remains similarly flexed but the

wrist experiences extension and slight ulnar deviation. Finally, at ball release is when the arm slot is measured, which is also a point of peak valgus torque at the elbow. During this time, the shoulder is adducted horizontally while the wrist is flexed and the elbow is pronated. The back leg leaves the ground with the stride leg temporarily bearing all the weight.

With an established risk of UCL injury associated with increased torque at the elbow, it was found by Camp et al. (2017) that varus elbow torque increased significantly with external shoulder rotation during the cocking phase of the pitch. However, there was also a significant relationship between elbow varus torque and arm rotation (cocking phase, $\chi 2 = 1392$, $p < .001$), arm speed (highest at acceleration phase, $\chi 2 = 57{,}683$, $p < .001$), and arm slot (varies by person but most significant at ball release, $\chi 2 = 428$, $p < .001$). With each Nm increase in varus torque, an athlete also experienced a 13° change in arm slot, a 116 deg/s increase in arm rotational speed, and an 8° increase in arm (shoulder external) rotation. Therefore, the authors also hypothesized that the interrelated nature of these parameters suggest that they should be collectively considered with respect to injury risk and treatment.

Though many position players will throw the ball, it is the main role of the pitcher position who, at the collegiate level, have been documented to throw 628 ± 476 pitches per season (Shanley et al., 2015). However, as high pitch count is associated with pain and injury in the upper extremity, it has been recommended that youth pitchers (≤12 years of age) throw no more than 70 pitches per day (Matsuura et al., 2021) and adolescent pitchers (ages 13-16) throw fewer than 96 pitches (Erickson et al., 2017). Further, it has been documented that when youth pitchers go on to play professional baseball, up to 50% of those with a history of exceeding pitch count recommendations are more likely to require UCL reconstruction surgery; “Tommy John” surgery (Erickson et al., 2017). As pitchers reach the professional level, the difference in pitch

count between those who experience injury requiring Tommy John surgery (2804 pitches per season) and those who do not (2826 pitches per season) is not statistically significant (Bakshi et al., 2019).

These findings suggest that perhaps disc golfers with mature skeletal anatomy are not susceptible to the same amount of risk as youth disc golfers, as disc golfers sometimes employ an overhead throw (Nelson et al., 2015) with similar mechanics to that of a baseball pitch. Nelson and colleagues (2015) found that 1.1% of disc golf injuries were attributed to overhead throws and, out of all mechanisms of injury, repetitive stress over time was responsible for roughly 57.4% of respondents' injuries. However, a higher proportion of disc golf injuries (12.7%) were found in those using forehand throws, which is better compared to the sidearm throw commonly employed in baseball and other sports like handball (Skejø et al., 2019).

Injuries, Sidearm Throwing

Valgus stress experienced by the elbow during throwing has been established as a risk factor for UCL injury because, during the ball acceleration phase of the throw, the force can exceed the 64 Nm tensile strength of the UCL (Solomito et al., 2015). The consequences of throwing sidearm are magnified as contralateral trunk flexion is decreased, which increases valgus elbow torque. As a result, following many repetitions, the joint can be made unstable and weak, with increased humeroulnar joint space due to ligamentous laxity at the UCL which can subsequently contribute to further soft tissue injury or ulnar nerve damage. However, there is risk beyond UCL injury in throwing sidearm. Another condition that is associated with repetitive high valgus stress over time at the elbow arises from the compressive forces generated on the medial surface of the elbow. Called valgus extension overload syndrome, there can be damage to the olecranon process or trochlea of the elbow joint that generates bone spurs (osteophytes)

which cause pain and often require surgical treatment. These findings are in alignment with the Nelson et al. (2015) finding that elbow injuries, specifically those due to repetitive stresses over time, are the most commonly suffered injury in disc golfers.

Swinging Biomechanics

By far, the most common style of disc golf throw and therefore the throw employed during the occurrence of most injuries is the backhand throw (86.2%) (Nelson et al., 2015). In evaluating the backhand throw, the biomechanics seem to more closely resemble those employed by a sport that requires swinging. A “backhand” throw “utilizes motion of the shoulder and elbow while simultaneously shifting weight from the back to front leg and twisting of the trunk” (Nelson et al., 2015). As such, a comparison of sorts can be drawn to a two-handed backhand swing in tennis and a backhand disc golf throw. While disc golf backhand throws tend to be performed with an athlete’s dominant hand, one difference is that the two-handed backhand swing of a tennis racket, depending upon the volley to the player, is sometimes performed by the athlete's non-dominant hand as during this time, the dominant hand will be utilizing a forehand orientation (Loushin et al., 2022). An evaluation of the upper body biomechanics by Loushin et al. (2022) found that upon the backswing, the non-dominant elbow and wrist are flexed and pronated and the wrist is in a position of ulnar deviation. As the athlete accelerates the racket while the swing takes place, wrist and elbow orientation do not change (Loushin et al., 2022) but the shoulder, previously internally rotated and adducted, abducts horizontally and rotates externally (Busuttil et al., 2022). Following impact of the racket with the ball, the elbow and wrist extend and the forearm supinates (Loushin et al., 2022).

Similar mechanics translate to table tennis, where a Bańkosz et al. (2020) study found some slight biomechanical differences between biological sexes. The authors hypothesize that

sexual anatomical differences account for the finding that men use shoulder flexion and supporting movements in the lower body to help generate rotation through the torso, especially the chest. In contrast, women tended to employ greater use of shoulder supination, elbow extension, and wrist supination during impact with the ball. These findings included the tendency for men to generate a higher force on the ball due to a larger proportion of muscle mass.

A study by He et al. (2020) focused on the mechanics of the lower body of table tennis players. It had previously been found that lower trunk axial rotation allows a greater amount of energy transfer from the pelvis to the shoulder of the racket arm that will impact the ball. During the backswing, knee flexion, external rotation and abduction are employed while the ankle is internally rotated and inverted. During the forward swing, the pelvis rotates axially while the knee extends and rotates, the hip rotates and the ankle rotates externally and plantar flexes. These lower-body movements are less consistent with disc throwing than their upper-body counterparts. In the absence of studies specific to disc golf or frisbee throwing, a comparison can be made to the lower body mechanics of swinging a baseball bat. Before a swing, the batter stands with their hips perpendicular to their target and takes a short stride with the lead foot while their weight is placed on the trail leg (Orishimo et al., 2023). Immediately following the foot contact with the ground, the pelvis rotates toward the stride foot as the weight is shifted to the stride leg, with the batter's hips ending parallel to their target. Baseball swinging biomechanics of the lower body are slightly more similar to disc golf than an action like discus throw, since discus involves the thrower performing a revolution with their body prior to release of the disc (Liu and Yu, 2021).

Injuries, Swinging

Of studies evaluating injuries during swinging sports, many focus on golf swings. These can be evaluated as a plausible comparison to disc golf as the lower body mechanics employed during a disc golf throw closely mimic that of a baseball swing, and baseball swinging mechanics are noted to be similar to that of golf swings by Orishimo et al. (2023). Golfers tend to experience lumbar spine injuries as their most frequently injured joint, with lead hip and lead knee injuries also being common (Kanwar et al., 2021). This likely results from the loading at the lumbar spine while axial twisting and lateral flexion takes place. In vitro studies have found that deviation of the spine from its neutral position causing even moderate compression of the intervertebral disc causes predictable herniation of the disc to a degree that is dependent on the compressive load in combination with axial rotation and lateral flexion. It is not presently clear whether these actions have the same potential consequences in vivo. It is an expected possibility that load can be distributed from the lower back to be dispersed across distal joints like the hip and knee.

Due to the hip serving as a pivot point, it experiences rotational velocity and loads during external rotation and axial loading (Kanwar et al., 2021). Hence, the hip is also known to experience a variety of disorders associated with golf swings. These include

The kinetics of the rest of the body also place stresses on the knee joints during golf swings (Kanwar et al., 2021). For instance, hip internal rotation contributed to decreased knee flexion with simultaneous increase in knee abduction that are positively associated with increased risk for patellofemoral pain and ACL injury. During a golf swing the knee, especially the lead knee, experiences high loading as the whole-body center of mass shifts with the lead foot remaining planted. As such, following many repetitions, the knee is known to experience

osteoarthritis at the medial and lateral components of the knee and tibia femoral pain and localized injury and pain of the soft tissues nearby.

Summary

With a limited number of disc golf-specific studies having taken place to date, it is beneficial to have the Nelson et al. (2015) study that attempts to find connections between types of throws and locations of injuries. However, there are limitations that require consideration in data analysis. First, the respondents were 92.9% male. It is already documented that body measurements that differ between the genders lead to differential risks for injuries (Sakaguchi et al., 2014). For instance, women are two to nine times more likely to incur knee injuries than men (Tan et al., 2019). Nelson and colleagues' (2015) study found the knee to be the fourth most commonly injured location and the most common reason for surgery in their male-dominated data pool. It is possible that if a greater number of female disc golfers were polled, there would be an increase in knee injury and knee surgery reporting. This may, however, be a proportionate representation of the gender differences in disc golfers as the PDGA reported that women accounted for 6-8% of disc golfers between 1999-2017 (Woods, 2019a).

In addition, the type of throw employed by players was skewed toward backhand (86.2%), with forehand (12.7%) and overhead (1.1%) throwers being disproportionately represented (Nelson et al., 2015). Risk for selection bias exists in that the survey's link was positioned on the PDGA website. This mechanism favors a pool of individuals who are computer-literate and willing to participate in survey studies. Finally, not all disc golfers are PDGA members. The sample size participating (883) was fairly large, but the vast majority of individuals who participated had experienced an injury (874). It is likely that their injury and the

marketing of the survey caused them to participate in the study as it is unlikely that, in the population as a whole, more than 81% of disc golfers incur injury while disc golfing.

As Nelson et al. (2015) found the site for the greatest number of disc golf injuries to be the elbow, it is worth noting that the van Trigt et al. (2023) finding may suggest a protective mechanism. Their result that a 20% maximal hand grip was not enough to decrease the humeroulnar gap, thus providing a protective effect to the UCL, was not to the maximum 50% recommended in other studies. Therefore, it cannot be definitively stated whether handgrip strength is or is not connected to protection against damage in the elbow ligaments during pitching. It is worth investigating whether this finding translates to disc golf in that increased handgrip strength may translate to decreased elbow torque during disc golf throws.

In either instance, the demands placed on the various joints of the body during disc golfing lead to injury most often reported at the elbow, shoulder, spine, knee and hip (Nelson et al., 2015). Part of the reason for the injury at these locations has to do with the anatomical design of these areas. The shoulder girdle, for instance, is composed of the humerus, clavicle, sternum, and scapula but neither of them participate in particularly strong articulations (Kadi et al., 2017). Instead, they are held in alignment by ligaments and muscles and tendons of the rotator cuff and this high range of motion translates to low stability of the joint overall. The knee supports the weight of the superior body, and is also strongly reinforced by ligaments and, to a lesser degree than the shoulder, nearby muscles (Hassebrock et al., 2020). There is also a sex disparity observed in the likelihood for knee injury, where females tend to have wider hips and therefore different angles of stresses placed on the knee, which increases the chance of injury (Tan et al., 2019). The spine is stronger or weaker depending on what its role is at each region (Frost et al., 2019). Commonly injured intervertebral discs occur at sites with spinal curvature like the

cervical spine, or in areas that support a large amount of weight in comparison to their cross sectional area, like the lumbar spine. The ankle joint is commonly injured due to its setup where the two bones of the lower leg are positioned on top of the talus and reinforced by ligaments (Golanó et al., 2014). Most commonly, a person is plantar flexing and inverting their ankle when they incur ankle sprains.

As there are no disc golf biomechanics studies to date, it is necessary to draw comparisons to other pre existing research in order to establish background information. With respect to the upper body, the backhand throw in disc golf can be compared to the actions of swinging a tennis racket (Loushin et al., 2022) or the later component of a discus throw (Liu and Yu, 2021). Forehand throwing of a disc has biomechanics and an arm slot that allow it to be compared to sidearm throwing as is utilized in baseball (Manzi et al., 2023). Finally, the lower body of a disc golfer is undergoing mechanics during the throw that are most similar to those to a person who is swinging a baseball bat or golf club (Orishimo et al., 2023).

Sports that employ the swinging motions of the upper limb utilized in disc golf throws regularly have a common finding; the elbow is often injured. Nelson et al. (2015) document this in disc golfers, while Solomito et al. (2015) find it in baseball throws. Swinging injuries in golfers tend to primarily be focused on the lumbar spine and knee. While the hypothesized causative mechanism for knee injury does not overlap with much of the mechanics of disc golf, the torsion experienced at the lumbar spine during a swing is replicated during disc golf throwing. With that said Nelson et al. (2015) reported that the knee and back are the fifth and third most commonly injured locations for disc golfers so there may be more overlap between the two sports' movements than was previously considered. Regrettably, Nelson and colleagues (2015) do not delineate between regions of the spine when reporting injuries.

Disc golf is currently underrepresented in research (Glattke et al., 2019). There are public health implications for understanding more about its biomechanics. While there are elite disc golfers who are highly competitive, it is also a good way for individuals who are looking to enter a program of physical activity to have a way to be outdoors and to walk. It also inspires community and brings people together who are looking to stay healthy but cannot perform high impact exercise, including older people. There are also initiatives to make disc golf courses more accessible to the visually impaired, who are often at risk of obesity as many forms of physical activity are not safe for them. As the sport continues to grow and reach athletes of varying physiology, it becomes increasingly important that more research be performed to establish the biomechanics of disc golf as an individual entity rather than continuing to compare it to other sports. Further, physicians should be clinically aware of the injuries that disc golfers and experience, personal trainers should be equipped with information on how to best train disc golfers to avoid injuries so they can stay active, and physical therapists and similar diagnosticians should be aware of the injury pathology to integrate into their therapy plans.

CHAPTER THREE: METHODS

Overview

One of the fastest growing sports is Disc golf, yet there is a limited amount of scientific research studying the topic. Disc golfers tend to experience elbow, shoulder, back, and knee injuries, yet there are no studies that specifically investigate whether biomechanical form contributes to these issues (Nelson et al., 2015). This study will attempt to discern if there is a correlation between biometric parameters (independent variable), biomechanics, and/or outcomes in the form of disc distance traveled (dependent variable). Experienced disc golfers who are participating in tournaments will be recruited for participation where elbow torque (dependent variable) will be measured by a motus sleeve during three forehand throws and three backhand throws, after collecting biometric data including grip strength, body mass index, and body fat percentage. The statistical significance between lean body mass and disc distance as well as grip strength of experienced disc golf players and amount of torque experienced by the elbow joint and the amount of torque experienced by the elbow joint in experienced disc golf players and disc distance traveled during disc golf throws will be evaluated.

Design

Disc golf is a sport that was invented in the 1970s and is rapidly growing. In the United States alone, it has been played by millions of people. With the estimates at 12 million people and 500,000 regular players in 2015 (Nelson et al., 2015), it has only grown in popularity since then in the presence of social media (Woods, 2019a). As with anything regarding participation in athletics, the likelihood of injuries increases in individuals participating in disc golf. In a Nelson et al. (2015) study, 81% of surveyed disc golfers (n=883) reported that they had sustained an injury while disc golfing. Common injuries include elbow (n=325), shoulder (n=305), back

(n=218) and knee (n=199) and many injuries require surgery (n = 115). Of these injuries, nearly 95% were in individuals who had at least one year of experience disc golfing and 92.9% were in males (Nelson et al., 2015). With the three main injuries (elbow, back, shoulder) pertaining to the upper body, and despite the fact that the sport is growing, there is presently no scientific research on the body mechanics of the various styles of throws used while disc golfing. This descriptive study aims to analyze the torque placed on the joints of male and female adult disc golfers in an effort to add research to the field, increase awareness of "safe" biomechanical form, and improve understanding of the potential risks associated with disc golfing.

During data collection, disc golfers were observed throwing three forehand and three backhand throws as these are the most common overall throwing styles, at 86.2% and 12.7%, respectively, with the majority of elbow injuries sustained by forehand throws (p=0.014) (Nelson et al., 2015). By observing the disc golfers without intervention, it allows measurement of elbow torque, distance thrown, and velocity without manipulating any of a person's usual biomechanics. In addition, performing actions that are familiar to an individual decreases the likelihood of injuries which increases likelihood of participation and promotes an increased data pool. Conducting the study on disc golfers who are participating in a tournament ensures a degree of experience and decreases the likelihood that participants are still developing their throwing mechanics and have variability in their throws.

Research Questions

RQ1: Is there a connection between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player?

RQ2: Does increased grip strength in experienced disc golf players translate to decreased amount of torque on the elbow joint during disc golf throws?

RQ3: Does the elbow joint of experienced disc golfers who throw a further distance experience higher amounts of torque?

Hypotheses

The null hypotheses for this study are:

$H_0$1: There is no statistically significant relationship between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player.

$H_0$2: There is no statistically significant relationship between grip strength of experienced disc golf players and amount of torque experienced by the elbow joint.

$H_0$3: There is no statistically significant relationship between the amount of torque experienced by the elbow joint in experienced disc golf players and disc distance traveled during disc golf throws.

Participants and Setting

The participants for the study were selected by convenience sampling and included n = 6 female and n = 35 male disc golfers from ages ranging from 21-66 years registered to compete in disc golf tournaments in Northwestern Pennsylvania and Southwestern New York. They had an average of 9.4 years of experience with a range from 0.75-40.0 years and an average Professional Disc Golf Association (PDGA) rating of 888.2 with a range from 635-1021. Seven participants were not registered with the PDGA and so did not have a rating. Because tournaments accept a limited number of participants, the potential sample pool is relatively small and therefore a few different sampling dates occurred. A priori power analysis was conducted using G*Power (version 3.1.9.7;Heinrich-Heine-Universität Düsseldorf, Düsseldorf, Germany) as described by Kang (2021). For sample size estimation, effect size was set to 0.5, alpha= 0.05, power = 0.90,

based off of a similar biomechanical study using the motus sleeve in baseball pitchers (Kobayashi et al., 2023). With t-test correlation model design, the result was a sample size of 28 total disc golfers required for statistical significance.

Instrumentation

During the consent process, participants completed the Physical Activity Readiness Questionnaire (PAR-Q) to demonstrate eligibility in the study (Warbuton et al., 2011). Other data measured prior to throwing trials included height in cm measured by stadiometer (Seca, Inc.), maximal voluntary isometric contraction (MVIC) handgrip strength in kg measured using a hand grip dynamometer (Jamar, Inc.; calibrated according to manufacturer's guidelines) and body composition analysis (percentage) by a handheld bioelectrical impedance analysis (BIA) device (Omron, Inc.; calibrated according to manufacturer's guidelines). Distance thrown was measured using a laser range finder (Chain Finder; Disc Store, Omaha, NE). The valgus torque on participants' elbow joints was measured in Nm using a motus kinetic sleeve (Motus; Motus Global, Rockville Center, NY) worn by the participant during their throws. The motus is a wearable sensor that contains one 3-axis accelerometer and one 3-axis gyroscope and can be placed within clips, sleeves, or shirts to show rotation at the joint to which it is applied in live time (Gulledge et al., 2020). The sensor utilizes accelerometers to transmit information via Bluetooth to its smartphone application, PULSEthrow (v. 8.8.65; Driveline Baseball) (Kobayashi et al., 2023). This tool has been applied in other studies to athletes such as baseball pitchers to reduce injury, having gained information about the risks in the kinematics of their motion (Carroll and Hansen, 2019; Lizzio et al., 2020a; Lizzio et al., 2020b; Makhni et al., 2018; Okoroha et al., 2018a; Okoroha et al., 2018b; Okoroha et al., 2019). Participants wore a sleeve size which fit such that the device was placed proximal to the medial epicondyle of the humerus

on the medial surface of the arm and the sensor did not move while throwing. While the device can also collect information about arm slot, arm speed, and shoulder rotation, data on elbow valgus stress was used for the present study.

Procedures

After Institutional Review Board (IRB) approval was gained, participants volunteered for participation in-person based on interest after the study had been advertised on the Lake Erie Disc Golf Club Facebook page and eligible “walk-ups” were taken on location. Participants deemed eligible were a minimum of 18 years of age and participants in the tournament, as opposed to spectators.

On the day of the study, consent was completed, participants had any questions answered, and any participants that did not have a fully negative PAR-Q+ received counseling regarding their PAR-Q+ results. Once prepared to participate in the study, participants’ age, height, weight, and gender were obtained in order to calculate BMI. Bioelectrical impedance analysis (BIA) was used to gain information about body composition as it is known to be a safe, fast, and unobtrusive mechanism in its use (Duan et al., 2021). The final pre-throw measurement was the handgrip MVIC of their dominant throwing hand. Three trials were conducted by the standard method; handgrip dynamometer (McGrath et al., 2021). They were then fitted with a sleeve with the motus sensor placed over the medial epicondyle of the humerus on their dominant throwing arm. Participants used researcher-provided discs (Destroyer; Innova Champion Discs, Inc.) to perform three backhand throws and three forehand throws, each released within a measured distance to control for the velocity that can be gained during the "wind up” process. Markers were manually placed where the disc landed, to minimize the likelihood that the discs gained any “bounce and roll” distance. Between forehand and backhand sets, the distance was measured

using a laser rangefinder (measured in feet, converted to meters for statistical analysis) as was described by Suh et al. (2019). The discs were returned for the participants to throw the same three discs for the opposite sets.

Data Analysis

In order to test the relationships between parameters including lean body mass (LBM) and disc distance traveled, a Pearson correlation was performed (Field, 2018). As disc golfers each performed three forehand and three backhand throws, the furthest distance in each type of throw was used to perform the correlation. These parameters were referred to as “maximum backhand distance” and “maximum forehand distance” and were measured in feet (ft). To determine LBM in the field, body fat percentage was measured (BIA) and subtracted from the participant’s measured weight.

To test the relationship between grip strength and torque at the elbow, a Pearson correlation was performed (Field, 2018). The average of the three handgrip MVIC trials was used for the analysis, referred to as “average handgrip MVIC” and measured in pounds (lbs) and converted to the metric equivalent. The reason an average was used for handgrip MVIC is because to give a maximal effort on each trial, it slightly fatigues the muscles, so the early trials can perceivably impact the outcome of the later trials. An average was not used for distances or torque because a best effort was sought for these parameters. In addition, despite being a maximal effort, each concurrent throw is not hypothesized to fatigue and impact later throws to a meaningful degree due to the nature of the movement involving the full body. Instead, to determine which trial’s torque to use for statistical analyses, the torque calculated by the Motus sensor on the trial of the maximum backhand distance and maximum forehand distance were

used, respectively. These are referred to as “torque - maximum backhand distance” and “torque - maximum forehand distance” and were measured in Newton meters (Nm).

The same parameters were used to test the relationships between elbow torque (“torque - maximum backhand distance”, “torque - maximum forehand distance”) and disc distance traveled (“maximum backhand distance”, “maximum forehand distance”), using a Pearson correlation (Field, 2018). All data was recorded then deidentified and inputted into a database software program (Excel, Microsoft Office, Redmond, WA). All statistical analyses were performed using Statistical Package for Social Science (IBM SPSS Statistics for Windows, Version 23.0. Armonk, NY).

CHAPTER FOUR: FINDINGS

Overview

Parameters tested included biometric tests including LBM and handgrip MVIC, as well as performance measures including maximum backhand distance, maximum forehand distance, torque on maximum backhand distance, and torque on maximum forehand distance. Assumptions tested using a Shapiro-Wilk test found that the biometric and distance data followed a normal distribution (LBM, maximum backhand distance, maximum forehand distance, handgrip MVIC). The data regarding torque, however (torque on maximum backhand distance, torque on maximum forehand distance), deviated from normal distribution. These trends were reflected in the parameters' respective tests of skewness and kurtosis. A two-tailed Pearson correlation was run as well as a linear regression and found that while some relationships existed, no correlations were statistically significant and likely due to chance. Therefore, we failed to reject the null hypotheses.

Research Questions

RQ1: Is there a connection between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player?

RQ2: Does increased grip strength in experienced disc golf players translate to decreased amount of torque on the elbow joint during disc golf throws?

RQ3: Does the elbow joint of experienced disc golfers who throw a further distance experience higher amounts of torque?

Hypotheses

The null hypotheses for this study were:

$H_0$1: There is no statistically significant relationship between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player.

$H_0$2: There is no statistically significant relationship between grip strength of experienced disc golf players and amount of torque experienced by the elbow joint.

$H_0$3: There is no statistically significant relationship between the amount of torque experienced by the elbow joint in experienced disc golf players and disc distance traveled during disc golf throws.

The alternative hypotheses for this study were:

H_A1: Disc golf players with higher amount of lean body mass will have a further disc distance traveled compared to throws made by a player with comparatively low amount of lean body mass.

H_A2: Players with higher grip strength will experience a higher amount of torque at the elbow joint than players with a comparatively lower grip strength.

H_A3: Players with a higher amount of torque experienced at the elbow joint will have a further disc distance traveled compared to throws made by a player with comparatively low amounts of torque at the elbow.

Descriptive Statistics

A total of n = 41 (n = 35 male; age 36.9 ± 11.2 years; height 179.2 ± 5.2 cm; weight 93.0 ± 21.9 kg) (n = 6 female; age 35.0 ± 6.7 years; height 64.9 ± 1.71 cm; weight 75.0 ± 19.5 kg) disc golfers were consented and biometric information was gathered. As 7 participants were lost to attrition, 34 completed throwing trials. Of those, equipment failure prevented a body composition analysis on 6 individuals. As such, n = 28 for research question 1 investigating lean

body mass, and n=34 for research questions 2 and 3 which centered around throwing performance.

Overall, the participants had an average of 9.4 years of experience with a range from 0.75 - 40.0 years and an average age of 36.5 which ranged from ages 21-66. The majority were right-handed (n = 37); with only a few left-handed (n = 4). Their average height was 176.8 cm which ranged from 162.6 – 188.0 cm, with a median of 177.5 cm. The body weight averaged 92.40 ± 22.70 kg ranging from 51.71 – 153.20 kg, with a median of 97.80 kg. As such, participants' BMI averaged 29.6 ± 6.68, ranging from 17.1 – 48.8 with a median of 28.1. Body fat percentage averaged 27.3% ± 7.28% ranging from 13.6% - 42.5% with a median of 26.8%. Lean body mass averaged 68.1 ± 2.39 kg with a range from 45.9 to 91.8 kg.

Results

Assumption Tests

The first research question investigated whether there is a connection between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player. Using N = 28, a Shapiro-Wilk test was used to test normality and found that the lean body mass data fit a normal distribution (W = 0.970, p = 0.592) (table 8, appendix A). The backhand distance thrown also followed a normal distribution (W = 0.971, p = 0.605), as did the data for forehand throws (W = 0.982, p = 0.885) (table 9, appendix A).

Research question 2 was investigating whether increased grip strength in experienced disc golf players translates to a decreased amount of torque on the elbow joint during disc golf throws. The Shapiro-Wilk test of normality found that neither the data for backhand torque (W = 0.773, p = 0.000) (table 16, appendix A) nor forehand torque (W = 0.838, p = 0.000) (table 17,

appendix A) followed a normal distribution. The handgrip MVIC data, however, did fit a normal distribution (W = 0.960, p = 0.239) (table 17, appendix A).

Research question 3 investigated whether the elbow joint of experienced disc golfers experienced a higher amount of torque with increased disc distance traveled during throws. A Shapiro-Wilk test found a normal distribution for maximum backhand distance (W = 0.983, p = 0.871) (table 24, appendix A) and forehand distance (W = 0.982, p = 0.843) (table 25, appendix A). Neither backhand maximum torque (W = 0.773, p = 0.000) (table 24, appendix A), nor forehand maximum torque (W = 0.838, p = 0.000) (table 25, appendix A) were normally distributed.

Statistical Analyses

Research Question 1

Analysis showed that LBM had an average LBM of 68.29 ± 2.39 with a range of 45.92 (45.85 – 91.77) kg (Table 1; Table 4, appendix A).

Table 1

Measures of central tendency for lean body mass, maximum backhand distance, and maximum forehand distance

Measures of Central Tendency - RQ1					
	N	Minimum	Maximum	Mean	Median
Lean Body Mass (kg)	28	45.85	91.77	68.29	63.34
Max Backhand Distance (m)	28	37.49	153.00	95.77	94.48
Max Forehand Distance (m)	28	23.47	155.14	80.45	78.48

The data were skewed slightly left (-0.08 ± 0.44) with a platykurtic distribution (-0.70 ± 0.86) (table 6, appendix A). Maximum backhand distance had an average of 95.77 ± 5.87 m and a

range of 115.51 (37.49 – 153.00) m. The data were skewed slightly right (0.27 ± 0.44) with a slight leptokurtic distribution (0.62 ± 0.86). Maximum forehand distance had an average distance of 80.45 ± 5.42 m a range of 131.67 (23.47 – 155.14) m (table 1). The data had a slight positive skew (0.27 ± 0.44) with a leptokurtic distribution (0.62 ± 0.86).

A two-tailed Pearson correlation was conducted and the correlation coefficient (R = 0.029) ($R^2 = 8.298^{-4}$, figure 6, appendix A) indicates no statistically significant relationship between lean body mass and disc distance traveled on backhand throws (p = 0.884) (table 9, appendix A). An F-test was conducted to compare the variances between lean body mass and backhand distance. The test yielded an F-value of 0.22; p = 0.00. Regarding forehand throws, there was a positive association to LBM (R = 0.334) ($R^2 = 0.112$, figure 7, appendix A) that was not statistically significant (p = 0.082) (table 11, appendix A). An F-test conducted to compare variances between lean body mass and forehand distance yielded an F-value of 0.24; p = 0.00. Thus, there is not sufficient evidence to reject the null hypothesis that there is no statistically significant relationship between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player.

Research Question 2

Torque on the maximum backhand distance averaged 27.00 ± 4.18 Nm with a range of 110.00 (4.00 – 114.00) Nm (table 2; table 14, appendix A).

Table 2

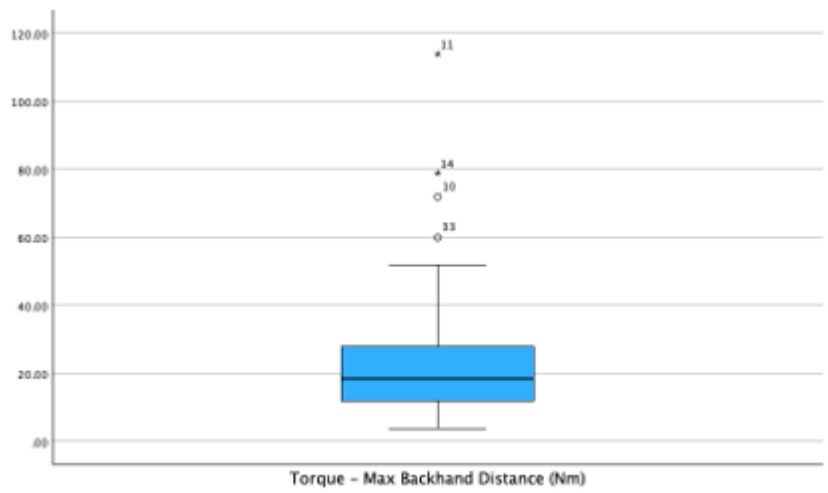
Measures of central tendency on average handgrip MVIC, torque on maximum forehand distance, and torque on maximum backhand distance

Measures of Central Tendency - RQ2					
	N	Minimum	Maximum	Mean	Median
Torque - Maximum Backhand Distance (Nm)	34	4.00	114.00	27.00	18.50
Torque - Maximum Forehand Distance (Nm)	34	6.00	84.00	25.41	21.00
Average Handgrip MVIC (kg)	34	33.02	72.24	51.74	50.27

The data had an extreme right skew (1.97 ± 0.40) and excess kurtosis (4.15 ± 0.79). The data showed four outliers (figure 1).

Figure 1

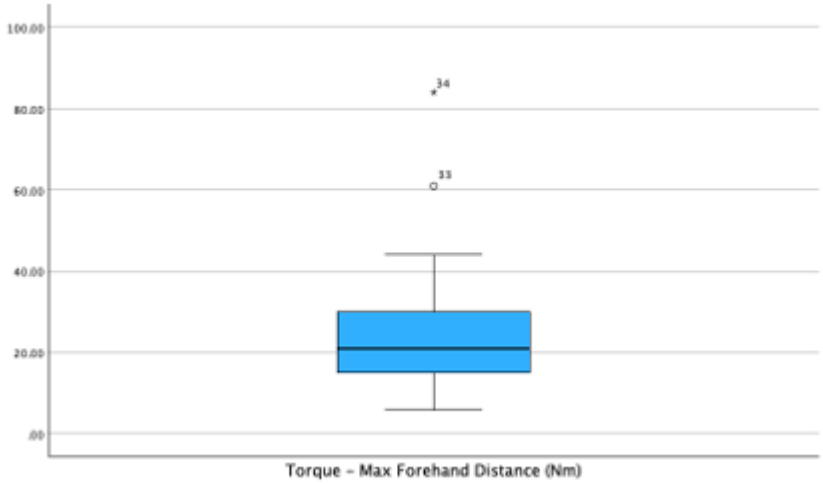
Torque on Maximum Backhand Distance showed four outliers: points 10, 11, 14 and 33, as shown below



Torque on the maximum forehand distance had a range of 78.00 (6.00 – 84.00) Nm (figure 2) with a mean of 25.41 ± 2.69 Nm (table 15, appendix A). The data showed extreme positive skewness (1.93 ± 0.40) with very high kurtosis (5.15 ± 0.79). The data showed two outliers (figure 2).

Figure 2

Torque on Maximum Forehand Distance contained two extreme outliers: points 33 and 34, as shown below



Average handgrip MVIC had a mean value of 51.74 kg and a range of 39.23 (33.02 – 72.24) kg (table 12, appendix A). Data skewed slightly positive (0.24 ± 0.40) with a slight platykurtic distribution (-0.27 ± 0.79).

A two-tailed Pearson correlation was conducted to examine the relationship between torque and handgrip MVIC. During backhand throws, the correlation coefficient (R = -0.04) (R^2 = 0.001, figure 11, appendix A) indicated a very weak negative relationship that lacks statistical significance between torque and handgrip strength (p = 0.84) (table 18, appendix A). An F-test was conducted to compare the variances between handgrip MVIC and backhand torque. The test yielded an F-value of 0.28; p = 0.00. With respect to forehand throws, a positive association exists (R = 0.148) (R^2 = 0.022, figure 12, appendix A) that is not statistically significant (p = 0.402) (table 19, appendix A). An F-test comparing variance between handgrip MVIC and forehand distance yielded an F-value of 0.44; p = 0.01. Thus, the null hypothesis that there is no statistically significant relationship between grip strength of experienced disc golf players and

amount of torque experienced by the elbow joint can be accepted. The observed correlations are likely due to random variation.

Research Question 3

Maximum backhand distance had an average of 98.91 ± 5.06 m and range of 115.51 (37.49 – 153.00) meters (table 22, appendix A).

Table 3

Measures of central tendency for maximum backhand distance and the torque on the throw, and maximum forehand distance and its torque

Measures of Central Tendency - RQ3					
	N	Minimum	Maximum	Mean	Median
Torque - Max Backhand Distance (Nm)	34	4.00	114.00	27.0000	18.50
Maximum Backhand Distance (m)	34	37.49	153.00	98.9118	101.04
Torque - Max Forehand Distance (Nm)	34	6.00	84.00	25.4118	21.00
Maximum Forehand Distance (m)	34	23.47	155.14	85.0440	89.30

The data were skewed slightly left (-0.03 ± 0.40) with a platykurtic distribution (-0.61 ± 0.79) (table 22, appendix A). Torque on the maximum backhand distance had an average of 27.00 ± 4.18 Nm with a range of 110.00 (4.00 – 114.00) Nm. The data had a large positive skew (1.97 ± 0.40) and very high kurtosis (4.15 ± 0.79). Maximum forehand distance averaged 85.04 ± 4.86 m and ranged from131.67 (23.47 – 155.14) meters (table 23, appendix A). The data had a slight negative skew (-0.06 ± 0.40) with a leptokurtic distribution (0.19 ± 0.79). Torque on the maximum forehand distance had a mean value of 25.41 ± 2.69 Nm and range of 78.00 (6.00 – 84.00) Nm. The data were skewed right (1.93 ± 0.40) with excess kurtosis (5.15 ± 0.79).

A two-tailed Pearson correlation indicates no statistically significant relationship between torque and backhand throwing distance ($R = 0.17$, $p = 0.33$) ($R^2 = 0.030$, figure 13, appendix A)

(table 26, appendix A). An F-test was conducted to compare the variances between backhand torque and backhand distance. The test yielded an F-value of 0.44; $p = 0.01$. There was also a weak positive correlation between forehand distance thrown and torque ($R = 0.221$) ($R^2 = 0.049$, figure 14, appendix A) that was not statistically significant ($p = 0.208$) (table 27, appendix A). The F- comparing the variances between forehand torque and forehand distance yielded an F-value of 3.22; $p = 0.00$. This suggests that there is no statistically significant relationship between the amount of torque experienced by the elbow joint in experienced disc golf players and disc distance traveled during disc golf throws; the null hypothesis can be accepted, and the observed correlations are likely due to chance.

CHAPTER FIVE: CONCLUSIONS

Overview

This study was novel as it addressed biomechanical and physiological measurements of disc golfers and how they affect throwing ability. The sample size for a medium effect size was met, and most parameters (LBM, handgrip MVIC, maximum backhand distance, maximum forehand distance) met criteria for normality. Measures for torque on both maximum backhand distance and maximum forehand distance did not follow a normal distribution. Where individuality, form, and throwing mechanics may be able to explain this, there were still important findings. Most notably, some measures for torque exceeded what has been determined to be the upper limit by studies regarding other sports, yet the thrower was uninjured. As it is predicted that the sport of disc golf will only continue to grow, there are many promising venues for further research to expand upon this study's findings.

Discussion

The purpose of this descriptive study was to evaluate uninjured and healthy (as determined by PAR-Q+ (Warburton et al., 2011)) disc golfers who had enrolled themselves in a tournament in order to search for commonalities in their throwing biomechanics.

Research Question 1

Research question 1 investigated whether there is a connection between lean body mass and disc distance traveled during disc golf throws made by the same experienced disc golf player. Assumptions testing found that all parameters (LBM, backhand distance, and forehand distance) followed a normal distribution ($W = 0.970$, $p = 0.592$) ($W = 0.971$, $p = 0.605$), ($W = 0.982$, $p = 0.885$), respectively (table 8, appendix A) (table 9, appendix A).

As could be expected following the results of previous studies (Hirsch et al., 2016) with participants of varying heights, weights, and genders, LBM in this investigation also had a wide range (45.92; from 45.85 – 153.00 kg) (table 6, appendix A). The average LBM was 68.29 kg (table 6, appendix A). Maximum backhand distance (37.49 – 153.00 meters) and maximum forehand distance (23.47 – 155.14 meters) had similar ranges (115.51m backhand, 131.37 forehand) and the average maximum backhand distance (95.77m) was higher than forehand (80.45m) (table 6, appendix A) (table 7, appendix A). When Hirsch and colleagues (2016) studied Division I track and field athletes, they found that throwers had the highest weight, lean mass, and body fat percentage when compared against athletes from all other track events. The Hirsch et al. (2016) throwers had a lean mass of 64.2 ± 11.7 kg. While the present study's LBM were calculated to be 68.29 ± 2.39 kg (table 7), the methodology for obtaining LBM should be aligned before directly conducting a comparison.

The results of this study did not follow the trend suggested by Hector and Phillips (2018) that a reduction in skeletal muscle mass will translate to reduced performance in elite athletes. A Pearson correlation found a weak positive relationship between LBM and maximum backhand distance ($R = 0.029$, $p = 0.884$, $R^2 = 8.298^{-4}$) (figure 6, appendix A) (table 7, appendix A) and maximum forehand distance $R = 0.334$, ($p = 0.882$, $R^2 = 0.112$) (figure 7, appendix A) (table 11, appendix A). While it can be said that disc golfing does not involve elite athletics and the sample size was not large, the disc golfer with the highest LBM turned out to have the lowest backhand distance thrown and among the lowest forehand distance (figures 6 & 7, appendix A).

Additionally, the players with the furthest maximum forehand distance throws were near the median values for lean body mass (figure 7) and the furthest throws for maximum backhand distance were distributed among athletes with varying LBM (figure 6). This may be able to be

explained by sample size and lack of prior research since this study is the first of its kind. However, technique and form may also be playing a role and would be a good venue for further research in the future.

Research Question 2

Research question 2 investigated if increased grip strength in experienced disc golf players translated to decreased amount of torque on the elbow joint during disc golf throws. This was partially informed by a Cronin et al. (2017) meta analysis which found a small relationship (7%) between handgrip strength and throwing in various sports (baseball, cricket, water polo, handball, American football, rugby, soccer, shot put, discus, javelin, hammer throw, bowling, basketball (shooting)), although it lacked statistical significance. While discus throwing was included, the authors suggested sport-specific testing and no other disc-based sports were evaluated that would allow a more direct comparison to disc golf.

In the present study, assumption testing found that while handgrip MVIC data fit a normal distribution ($W = 0.960$, $p = 0.239$) (table 17, appendix A). Average handgrip MVIC had a range of 39.23 (33.02 – 72.24) kg with a mean of 51.74 ± 1.79 kg (table 15, appendix A). To gain the proposed benefits to the ulnohumeral joint proposed by van Trigt et al. (2023), the thrower would have to activate over 20% of their handgrip MVIC over the course of the throw. Other studies (Tsubono et al., 2022) have suggested that this value may be closer to 50%. It is presently unknown whether disc golfers are gripping the disc at nearly half of their handgrip MVIC, and this could be a meaningful avenue for future research.

Neither the data for backhand torque ($W = 0.773$, $p = 0.000$) (table 16, appendix A) nor forehand torque ($W = 0.838$, $p = 0.000$) (table 17, appendix A) were normally distributed. It is believed that the maximum tensile strength of the UCL during sidearm baseball throws is 64 Nm

(Solomito et al., 2015). Three athletes exceeded 64 Nm on backhand throws (Figure 1), as torque on the maximum backhand distance had a range of 110.00 (4.00 – 114.00) Nm (table 14, appendix A) with an average of 27.00 ± 4.18 Nm. Only one athlete exceeded 64 Nm on forehand throws (Figure 2), as torque on the maximum forehand distance had a range of 78.00 (6.00 – 84.00) Nm (table 15, appendix A) with a mean of 25.41 ± 2.69 Nm. The null hypothesis that there is no statistically significant relationship between grip strength of experienced disc golf players and amount of torque experienced by the elbow joint was not rejected.

It is possible that since torque measurements in this study were taken by a sensor that communicated results to a smartphone application via Bluetooth (Kobayashi et al., 2023), these discrepancies could be due to measurement bias. However, because Solomito and colleagues (2015) were investigating baseball throwers, the fact that disc golfers exceeded this measure could be in differences of shape or weight of the projectile, or factors surrounding throwing form like arm slot or grip position.

Research Question 3

Research question 3 attempted to find whether the elbow joint of experienced disc golfers experiences a higher amount of torque with increased disc distance traveled during throws. This research question was formed, in part, based on the finding of a positive association between elbow varus torque and ball speed ($R^2 = 0.05$; $P < .001$) (Tanaka et al., 2020). In this present study, neither backhand torque ($W = 0.773$, $p = 0.000$) (table 24, appendix A), nor forehand torque ($W = 0.838$, $p = 0.000$) (table 25, appendix A) were normally distributed.

As $KE = 1/2mv^2$, and with the sampling conditions and mass of the discs held constant, it is assumed that increased velocity (speed) translates to increased disc distance traveled. It was found that maximum forehand distance had a range of 131.67 (23.47 – 155.14) meters, with an

average distance of 85.04 ± 4.86 m (table 23, appendix A). Torque on the maximum forehand distance had a range of 78.00 (6.00 – 84.00) Nm with a mean of 25.41 ± 2.69 Nm.

Both maximum backhand distance (W = 0.983, p = 0.871) (table 24, appendix A) and forehand distance (W = 0.982, p = 0.843) (table 25, appendix A) followed normal distribution. Maximum backhand distance had a range of 115.51 (37.49 – 153.00) meters, with an average of 98.91 ± 5.06 m (table 22, appendix A). Torque on the maximum backhand distance had a range of 110.00 (4.00 – 114.00) Nm with an average of 27.00 ± 4.18 Nm. Previous studies investigating baseball throws measured peak elbow valgus torque at 50 Nm (van Trigt et al., 2023), and peak elbow varus torque at 57.0 ± 15.7 Nm (Tanaka et al., 2020). These values were exceeded by the disc golfers, and whether this is due to mechanics, range of motion, or another phenomenon warrants further investigation.

Implications

To the present knowledge, this study is the first of its kind to assess disc golf from a biometric and performance standpoint. The sport of disc golf is growing rapidly in popularity (Woods, 2019a), and accessible to people of varying abilities and skill levels (Glattke et al., 2019). As such, it stands to be argued that physicians, personal trainers, and athletic trainers should be able to access evidence-based information about the risks to the musculoskeletal system both from a treatment perspective regarding injuries and from a perspective of performance enhancement.

In addition, while arm slot (the angle of the arm at release, relative to the ground) was not a variable that was investigated in the present study, it is a component that is measurable by the motus sleeve. If evaluated in future studies, it could be helpful for video tracking software to correlate form to arm slot angles throughout the duration of the throw. In the present study, raw

data from the motus readouts for each disc golf throw also included information about arm slot. During backhand throws, disc golfers averaged a 10.5° ± 16.7° arm slot with a range from -38.0 to 66.0°. For forehand throws, disc golfers' arm slot averaged 8.1° ± 24.5° with a range from -56.0 to 88.0°. This initiates the ability for comparison of disc golf to pre-existing literature from other sports which is vastly more common. For instance, Manzi et al. (2023) provided arm slot for overhand (37.1° ± 7.4°), sidearm (63.2° ± 6.6°) and three-quarter (49.7° ± 2.9°) baseball throws. This was an expansion on the Escamilla et al., (2018) finding regarding baseball pitchers that overhand throws had ≤40° arm slot angle, three-quarter ranges from 41° to 69°, and sidearm is ≥70°. While the measured range for disc golfers was still extremely wide (104° backhand, 144° forehand), the values place forehand throws in a position to draw comparisons to sidearm baseball throws based on arm slot angles. This has implications for using evidence-based practices for predicting injuries that a disc golfer may be susceptible to based on their form, which is currently impossible as there is no prior research on the subject.

Limitations

One limitation of the study was introduced by instrumentation in obtaining LBM. This sample size was reduced (n=28) as compared to the sample size for research questions 2 and 3 (n=34) because the Omron handheld bioelectrical impedance analysis (BIA) device was nonfunctional during those trials. Technological issues are unavoidable in research, but it reduced the sample size and required that those throws be excluded during statistical analysis.

Additionally, bioelectrical impedance analysis simply measures all body contents that return an electrical pulse (Campa et al., 2024). Therefore, readouts are impacted by non-muscular factors that may impact throwing distance like hydration status, as well as others that are not related to performance such as stomach contents and bladder volume. Fat mass was

extrapolated by measuring the person's clothed weight and body fat percentage by BIA. That is a flawed system because non-conductive weight-bearing items like the clothing, skeleton, and some organs will have been included with what was calculated as LBM. All people in the study were clothed and have skeletal and organ weight, but it cannot simply be said that we can evaluate individuals proportionately.

There should not be major differences in skeletal weight. The range in body weight was from 51.71 – 153.20 kg, with an average 92.40 ± 22.7 kg and median of 97.80. The skeleton reflects roughly 12-15% of a person's body weight (Borisov and Marei, 1974). This means that the skeletal weight of the participants at the bottom of the weight range may have been 6.20 – 7.76kg, while the top of the weight range would be roughly 18.38 – 22.98 kg. It is likely that more of this variability comes from soft tissue rather than skeletal weight, since the range in height was only 27.94 cm, from 162.6 – 188.0 centimeters overall with an average of 176.8 ± 7.3 cm and a median of 177.5 cm. There is presumably many body systems' individuality contributing to the variety in the results. Field research dictates that non-portable options like a dual energy X-ray absorptiometry (DEXA) scan or underwater weighing cannot be used, even though those are more accurate for determining body composition (Tornero-Aguilera et al., 2022). However, more sophisticated BIA devices exist than the one used in the present study that can specifically calculate skeletal muscle mass and that tool would be better suited for field measurements. It could be beneficial in future studies to introduce a second method to improve consistency. For instance, a plausible method for dual-measurement may be a body fat caliper.

There was also a small chance of interaction effects where the fit of the motus sleeve impaired the ability of the participant to throw with their best effort. The sleeve was worn by participants such that the proximal and distal elastic bands that hold the sleeve in place were

positioned away from any of the joints and the pouch that houses the sensor was positioned proximal to the medial epicondyle of the humerus. The motus sleeve is adapted to be used on athletes during exercise, so it is designed in a way that avoids impeding movement (Kobayashi et al., 2023). While it is unlikely that the fit of the device affected performance, the chances of some impact were not measured, so this factor should be considered.

Also unlikely is the risk of fatigue due to repeated measures. Since experienced disc golfers were selected, the risk is minimal. However, six throws were performed by each participant after they had been afforded ample throws to warm up. As such, some testing effects play a threat to internal validity of the study. There is also a chance that, with participants preparing to compete, that they did not truly give their best effort. As we wanted the throwing form to be as natural as possible, another limitation is that we did not control for finger placement on the disc (grip). This could impact the results, but we wanted the most comfortable form for the participants in order to ascertain the true maximum possible distance on each throw.

A limitation in the field that should be accounted for in future research is the tendency for discs to bounce or roll after landing, complicating the physical placement of the marker. To have an accurate reading of distance, an observer standing in the field must run to where the disc is about to land and place a distance marker as it hits to minimize the influence of bounce and roll in the reading. As it is not possible to be in two places at once, the researcher can only use the laser rangefinder or drop distance markers; not both. There was some risk to inter-rater reliability to have a different observer placing the markers between trials. There was also a limitation due to attrition. Some participants that were consented and had biometric data collected dropped out of the study in the interest of time, due to the trials consuming too much time with a single researcher dropping distance markers. In addition, disc golf tournaments commonly take place at

wooded areas. As such, there were some trials where discs were thrown into overgrown areas and in those instances, it was not possible to see if they bounced or rolled to where they were located.

Finally, climate and weather were potential confounding variables. A test that strictly measures performance would be better performed in a laboratory setting where humidity, wind, and temperature can be held constant as these parameters can all affect the disc and therefore, the throw (Giljarhus et al., 2022). During data collection, the participants were coached to give their best effort and many of them independently appeared to account for the wind speed and direction in doing so. Drag may been a variable force on the disc, however, participants threw the same discs between trials and each other, within a one-hour span of each other. This minimizes the likelihood that drag would greatly impact the results. As disc golf will continue to take place at outdoor venues, this may contain some threats to ecological validity. Trials conducted under controlled conditions would not accurately evaluate the sport in practicality.

Recommendations for Future Research

An intent of this study was to gain information about biometrics and biomechanics of uninjured disc golfers to establish a healthy baseline. As such, it would be ideal to have data from participants who were injured during throws as well. While injuries cannot be induced, a way to capture this type of data may be to outfit disc golfers with a motus sleeve and sensor for all of their throws and record data longitudinally. This could be performed alongside some type of journaling activity so throwers could document times when throws resulted in soreness or injury. While it would require a large research team to have someone present to accurately measure distance, this may not be necessary as disc golfers commonly carry a laser rangefinder

to evaluate their performance. Under these conditions, distance could be self-reported, although this format would leave room for error or bias.

With respect to adding to the biomechanical body of evidence for disc golf, another meaningful study could be a meta-analysis investigating what has been measured for elbow torque for each sport. As disc golf is growing in popularity but still not as well-known as sports like baseball, tennis, and football, the current best effort for drawing biomechanical comparison is in investigating what actions a disc throw is similar to. The present study suggests that forehand throws can feasibly be compared to sidearm baseball throws, as the palm of the hand faces forward during the release phase of both types of throw. While it was out of scope for this study which investigated the distance of a thrown projectile, a good comparison for a backhand throw would likely be a one-handed backhand shot in a sport that utilizes a stick or racket, like table tennis. Further research could also investigate the connection between biomechanics at each stage of the throw and performance.

It could also be an impactful study to investigate the third type and least common style of disc golf throw; overhead. This study did not investigate overhead throws as they are the least commonly utilized; about 1.1% of players' primary throws were overhead throws, compared to 12.7% forehand and 86.2% backhand (Nelson et al., 2015). Therefore, the forehand throw is not necessarily of the most immediate priority. However, as the sport grows in popularity there is certainly something to be gained in understanding the biomechanical stresses incurred during all types of throws.

Additionally, while the elbow is the most commonly injured joint in disc golfers, there are other joints of concern, such as the shoulder and knee (Nelson et al., 2015). Often, injuries to these joints can require surgery and/or impact the athlete's quality of life or ability to work at

their job. To begin addressing these issues, it would be advantageous to expand the study to more joints in order to establish a healthy baseline and could include parameters including rotational velocity of either or both the arm (shoulder joint) or forearm (elbow joint). This guidance aligns with the suggestion by Tanaka et al. (2020) that varus elbow torque was affected by the mechanics of the other joints contributing to the throwing motion. Additionally, the findings of Miyazaki et al. (2021) indicate that changes in joint flexion upon release of a baseball throw can affect mechanics within the kinetic chain, potentially increasing torque on other joints and the subsequent rate of injury. As the shoulder is the second most commonly injured joint in disc golfers (Nelson et al., 2015), that would be a logical next location for investigation.

Another parameter that could be evaluated in future studies is the effect (or lack thereof) of grip. It is hypothesized that grip strength can close the humeroulnar gap in baseball players and have a protective effect on the UCL (van Trigt et al., 2023). To find out if a similar effect translates to disc golf throws, grip should be controlled. To gain an understanding of disc golfers' opinions of this study, however, participants were asked between trials if they felt that they could throw their best effort if grip was controlled by the researchers. They all predicted anecdotally that if their grip were changed from what they are familiar with, that they could not throw their best distance. Finger position was not manipulated during this study to minimize the likelihood of multiple-treatment interference and ensure that maximum distance was gained. Finally, to collect data strictly at tournaments on experienced disc golfers was advantageous for this study because it decreased the chances that a participant would be injured. In addition, since tournaments require payment upon entry, the current study design contained some selection bias as it selected for individuals who have the socioeconomic means to participate. These individuals may also be more likely to have access to pre-habilitative measures like gym memberships and

healthcare such that they were less likely to be injured. It would be advantageous in the future to increase the size and diversity of the sample pool by also including individuals who are recreational disc golfers that would not enroll themselves in a tournament.

A limitation realized following data analysis includes the presence of outliers. Outliers existed in measures of torque on maximum backhand distance (n = 4; participants 10, 11, 14 and 33) and torque on maximum forehand distance (n = 2, participants 33 and 34). These likely contributed to the extreme skew (1.97 ± 0.40) (1.93 ± 0.40) and kurtosis (4.15 ± 0.79) (5.15 ± 0.79) values for torque on maximum backhand and forehand distance, respectively. In looking into the outliers' three trials, some interesting patterns emerge. Participant 10's backhand torque was reported as 114 Nm but their other two backhand trials experienced torque of 17 and 19 Nm. The other three outliers had torque values within a close range (50-51 Nm; 43-52 Nm; 50-60 Nm) suggesting that a high torque may be induced by their form. Two of the backhand outliers were two of four touring professional disc golfers that were sampled. One individual (participant 33) was also one of the two outliers for torque on maximum forehand distance (57-62 Nm). The other outlier on forehand torque was also a touring professional and the torque on their three trials were also all relatively high (72-84 Nm).

Perhaps the outliers are attributable to malfunction of the motus sensor. Future research could aim to develop a method (such as using two sensors for outputs and comparing values for consistency) to determine if equipment is malfunctioning. Future research could also look for trends within the outliers or conduct statistical analyses which remove outliers to improve robustness provided that the sample size is large enough to not lose statistical relevance with their exclusion.

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APPENDIX A

Table 4

Case Processing Summary; LBM vs. Maximum Backhand Distance (RQ1)

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Lean Body Mass (kg)	28	100.0%	0	0.0%	28	100.0%
Max Backhand Distance (m)	28	100.0%	0	0.0%	28	100.0%

Table 5

Case Processing Summary; LBM vs. Maximum Forehand Distance (RQ1)

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Lean Body Mass (kg)	28	100.0%	0	0.0%	28	100.0%
Max Forehand Distance (m)	28	100.0%	0	0.0%	28	100.0%

Table 6

Descriptive Statistics; LBM vs. Maximum Backhand Distance (RQ1)

Descriptive Statistics			
		Statistic	Std. Error
Lean Body Mass (kg)	Mean	68.29	2.39
	Minimum	45.85	
	Maximum	91.77	
	Range	45.92	
	Skewness	-0.08	0.44
	Kurtosis	-0.70	0.86
Max Backhand Distance (m)	Mean	95.77	5.87
	Minimum	37.49	
	Maximum	153.00	
	Range	115.51	
	Skewness	0.22	0.44
	Kurtosis	-0.62	0.86

Table 7

Descriptive Statistics; LBM vs. Maximum Forehand Distance (RQ1)

Descriptive Statistics			
		Statistic	Std. Error
Lean Body Mass (kg)	Mean	68.29	2.39
	Minimum	45.85	
	Maximum	91.77	
	Range	45.92	
	Skewness	-0.08	0.44
	Kurtosis	-0.70	0.86
Max Forehand Distance (m)	Mean	80.45	5.42
	Minimum	23.47	
	Maximum	155.14	
	Range	131.67	
	Skewness	0.27	0.44
	Kurtosis	0.62	0.86

Table 8

Tests of Normality; LBM vs. Maximum Backhand Distance (RQ1)

Tests of Normality						
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Lean Body Mass (kg)	0.111	28	0.200*	0.970	28	0.592
Max Backhand Distance (m)	0.077	28	0.200*	0.971	28	0.605

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 9

Tests of Normality; LBM vs. Maximum Forehand Distance (RQ1)

Tests of Normality						
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Lean Body Mass (kg)	0.111	28	.200*	0.970	28	0.592
Max Forehand Distance (m)	0.087	28	.200*	0.982	28	0.885

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 10

Pearson Correlation; LBM vs. Maximum Backhand Distance (RQ1)

Correlations			
		Lean Body Mass (kg)	Max Backhand Distance (m)
Lean Body Mass (kg)	Pearson		0.029
	Sig. (2-tailed)		0.884
	N		28
Max Backhand Distance (m)	Pearson	0.029	
	Sig. (2-tailed)	0.884	
	N	28	

Table 11

Pearson Correlation; LBM vs. Maximum Forehand Distance (RQ1)

Correlations			
		Lean Body Mass (kg)	Max Forehand Distance (m)
Lean Body Mass (kg)	Pearson		0.334
	Sig. (2-tailed)		0.082
	N		28
Max Forehand Distance (m)	Pearson	0.334	
	Sig. (2-tailed)	0.082	
	N	28	

Figure 3

Normal Distribution Curve; Maximum Backhand Distance (RQ1)

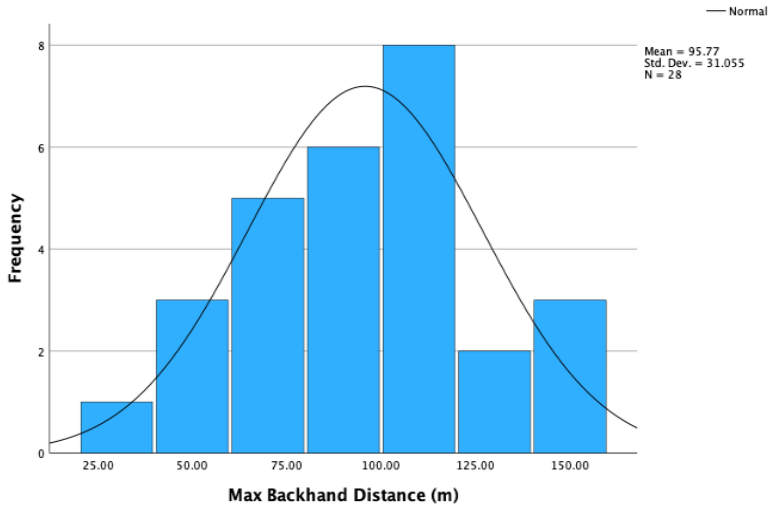


Figure 4

Normal Distribution Curve; Lean Body Mass (RQ1)

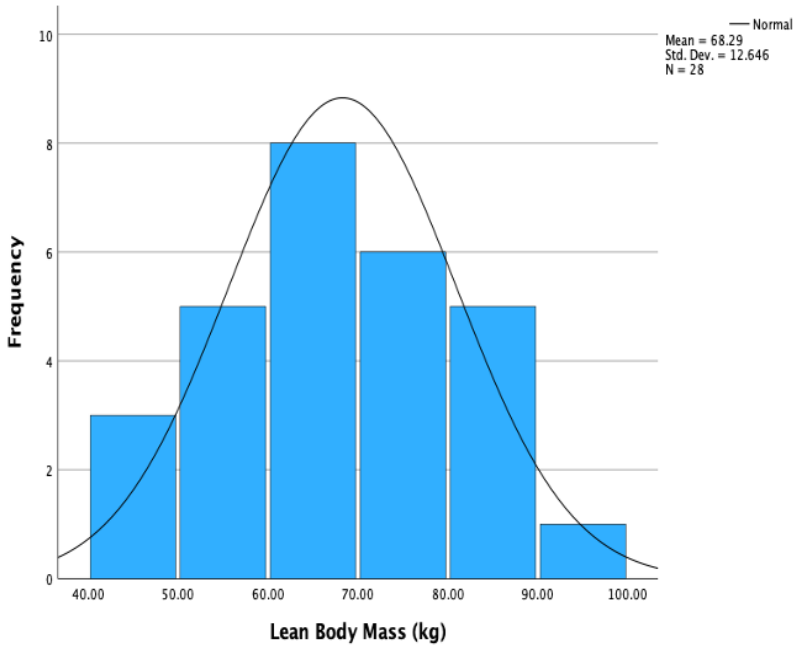


Figure 5

Normal Distribution Curve; Maximum Forehand Distance (RQ1)

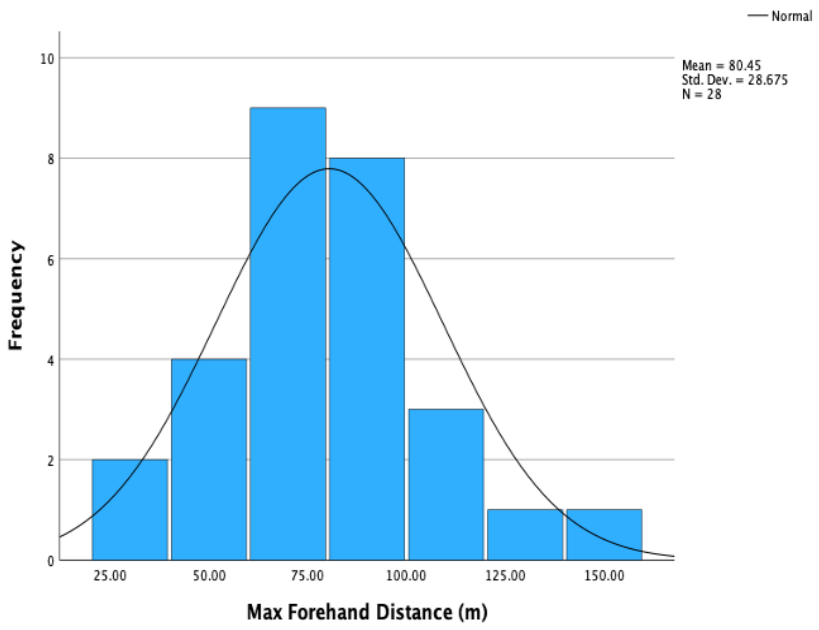


Figure 6

Scatterplot; LBM vs. Maximum Backhand Distance (RQ1)

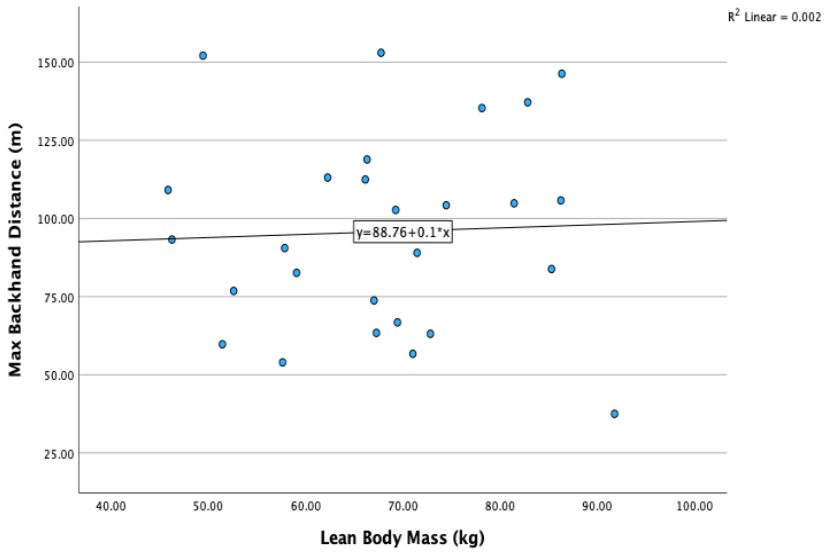


Figure 7

Scatterplot; LBM vs. Maximum Forehand Distance (RQ1)

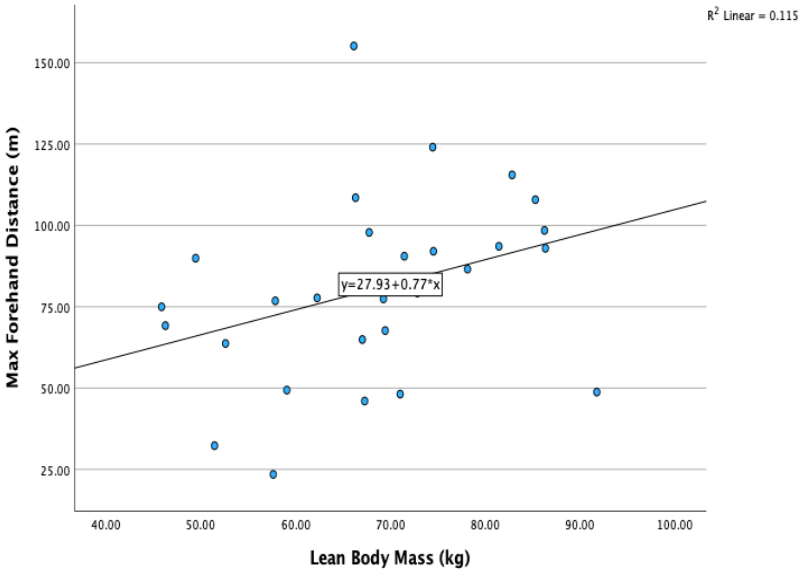


Table 12

Case Processing Summary; Average Handgrip MVIC vs. Torque - Maximum Backhand Distance (RQ2)

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Torque - Maximum Backhand Distance	34	100.0%	0	0.0%	34	100.0%
Average Handgrip MVIC (kg)	34	100.0%	0	0.0%	34	100.0%

Table 13

Case Processing Summary; Average Handgrip MVIC vs. Torque - Maximum Forehand Distance (RQ2)

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Average Handgrip MVIC (kg)	34	100.0%	0	0.0%	34	100.0%
Torque - Maximum Forehand Distance	34	100.0%	0	0.0%	34	100.0%

Table 14

Descriptive Statistics; Average Handgrip MVIC vs. Torque - Maximum Backhand Distance (RQ2)

Descriptive Statistics			
		Statistic	Std. Error
Torque - Maximum Backhand Distance (Nm)	Mean	27.00	4.18
	Minimum	4.00	
	Maximum	114.00	
	Range	110.00	
	Skewness	1.97	0.40
	Kurtosis	4.15	0.79
Average Handgrip MVIC (kg)	Mean	51.74	1.79
	Minimum	33.02	
	Maximum	72.24	
	Range	39.23	
	Skewness	0.24	0.40
	Kurtosis	-0.27	0.79

Table 15

Descriptive Statistics; Average Handgrip MVIC vs. Torque - Maximum Forehand Distance (RQ2)

Descriptives			
		Statistic	Std. Error
Average Handgrip MVIC (kg)	Mean	51.74	1.79
	Minimum	33.02	
	Maximum	72.24	
	Range	39.23	
	Skewness	0.24	0.40
	Kurtosis	-0.27	0.79
Torque - Maximum Forehand Distance (Nm)	Mean	25.41	2.69
	Minimum	6.00	
	Maximum	84.00	
	Range	78.00	
	Skewness	1.93	0.40
	Kurtosis	5.15	0.79

Table 16

Tests of Normality; Average Handgrip MVIC vs. Torque – Maximum Backhand Distance (RQ2)

Tests of Normality						
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Torque - Maximum Backhand Distance (Nm)	0.261	34	0.000	0.773	34	0.000
Average Handgrip MVIC (kg)	0.114	34	0.200*	0.960	34	0.239

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 17

Tests of Normality; Average Handgrip MVIC vs. Torque – Maximum Forehand Distance (RQ2)

Tests of Normality						
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Average Handgrip MVIC (kg)	0.114	34	0.200*	0.960	34	0.239
Torque - Maximum Forehand Distance (Nm)	0.166	34	0.019	0.838	34	0.000

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 18

Pearson Correlation; Average Handgrip MVIC vs. Torque - Maximum Backhand Distance (RQ2)

Correlations			
		Torque - Maximum Backhand Distance (Nm)	Average Handgrip MVIC (kg)
Torque - Maximum Backhand Distance (Nm)	Pearson		-0.04
	Sig. (2-tailed)		0.84
	N		34
Average Handgrip MVIC (kg)	Pearson	-0.04	
	Sig. (2-tailed)	0.84	
	N	34	

Table 19

Pearson Correlation; Average Handgrip MVIC vs. Torque - Maximum Forehand Distance (RQ2)

Correlations			
		Average Handgrip MVIC (kg)	Torque - Maximum Forehand Distance (Nm)
Average Handgrip MVIC (kg)	Pearson		0.148
	Sig. (2-tailed)		0.402
	N		34
Torque - Maximum Forehand Distance (Nm)	Pearson	0.148	
	Sig. (2-tailed)	0.402	
	N	34	

Figure 8

Normal Distribution Curve; Torque - Maximum Backhand Distance (RQ2)

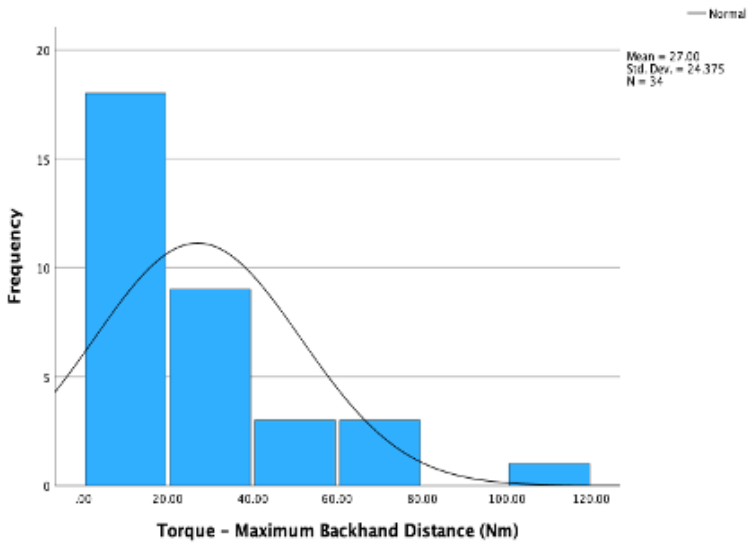


Figure 9

Normal Distribution Curve; Average Handgrip MVIC (RQ2)

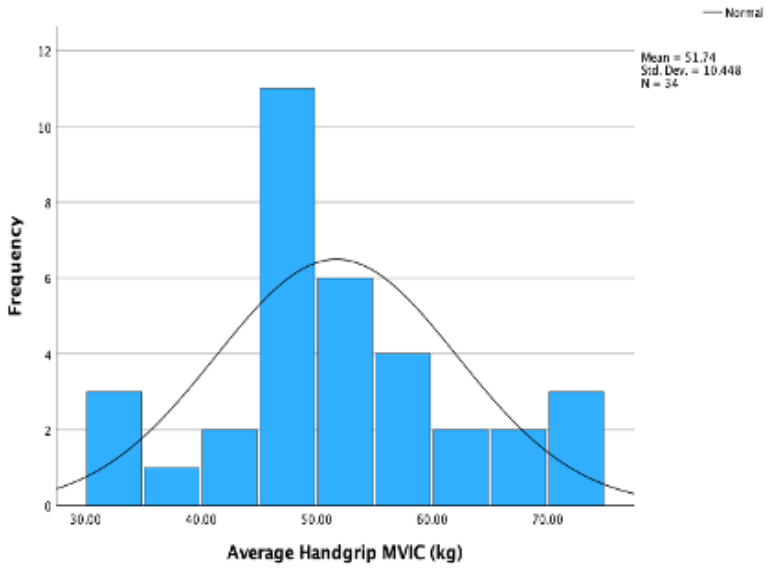


Figure 10

Normal Distribution Curve; Torque - Maximum Forehand Distance (RQ2)

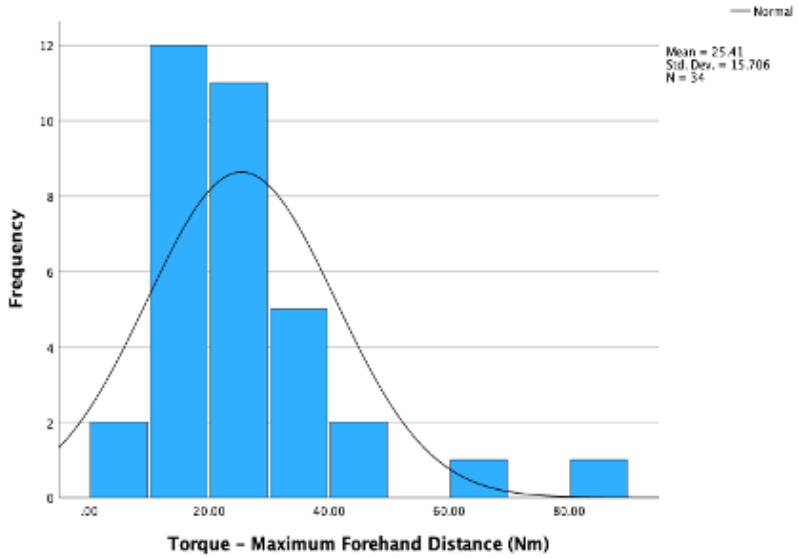


Figure 11

Scatterplot; Torque - Maximum Backhand Distance vs. Average Handgrip MVIC (RQ2)

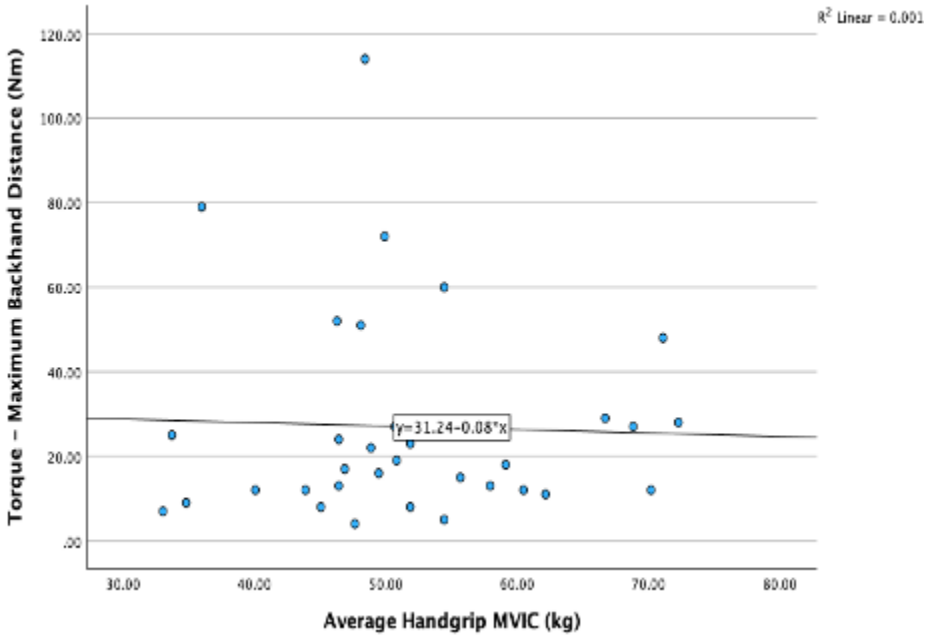


Figure 12

Scatterplot; Torque - Maximum Forehand Distance vs. Average Handgrip MVIC (RQ2)

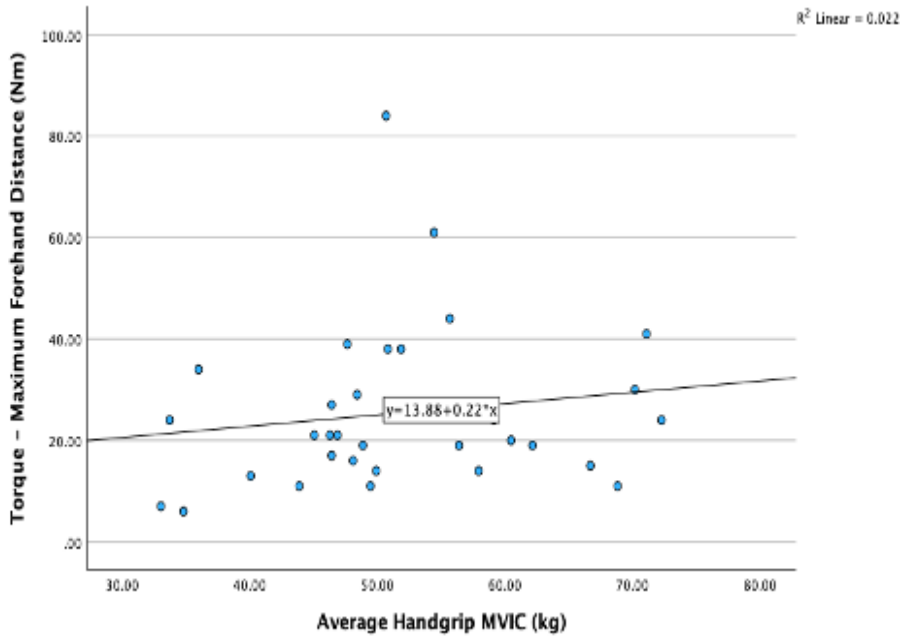


Table 20

Case Processing Summary; Backhand Maximum Distance vs. Torque (RQ3)

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Maximum Backhand Distance (m)	34	100.0%	0	0.0%	34	100.0%
Torque - Max Backhand Distance (Nm)	34	100.0%	0	0.0%	34	100.0%

Table 21

Case Processing Summary; Forehand Maximum Distance vs. Torque (RQ3)

Case Processing Summary						
	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Maximum Forehand Distance (m)	34	100.0%	0	0.0%	34	100.0%
Torque - Max Forehand Distance (Nm)	34	100.0%	0	0.0%	34	100.0%

Table 22

Descriptive Statistics; Backhand Maximum Distance vs. Torque (RQ3)

Descriptive Statistics			
		Statistic	Std. Error
Torque - Max Backhand Distance (Nm)	Mean	27.00	4.18
	Minimum	4.00	
	Maximum	114.00	
	Range	110.00	
	Skewness	1.97	0.40
	Kurtosis	4.15	0.79
Maximum Backhand Distance (m)	Mean	98.91	5.06
	Minimum	37.49	
	Maximum	153.00	
	Range	115.51	
	Skewness	-0.03	0.40
	Kurtosis	-0.61	0.79

Table 23

Descriptive Statistics; Forehand Maximum Distance vs. Torque (RQ3)

Descriptive Statistics			
		Statistic	Std. Error
Torque - Max Forehand Distance (Nm)	Mean	25.41	2.69
	Minimum	6.00	
	Maximum	84.00	
	Range	78.00	
	Skewness	1.93	0.40
	Kurtosis	5.15	0.79
Maximum Forehand Distance (m)	Mean	85.04	4.86
	Minimum	23.47	
	Maximum	155.14	
	Range	131.67	
	Skewness	-0.06	0.40
	Kurtosis	0.19	0.79

Table 24

Tests of Normality; Backhand Maximum Distance vs. Torque (RQ3)

Tests of Normality						
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Torque - Max Backhand Distance (Nm)	0.261	34	0.000	0.773	34	0.000
Maximum Backhand Distance (m)	0.068	34	0.200*	0.983	34	0.871

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 25

Tests of Normality; Forehand Maximum Distance vs. Torque (RQ3)

Tests of Normality						
	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Torque - Max Forehand Distance (Nm)	0.166	34	0.019	0.838	34	0.000
Maximum Forehand Distance (m)	0.089	34	0.200*	0.982	34	0.843

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Table 26

Pearson Correlation; Backhand Maximum Distance vs. Torque (RQ3)

Correlations			
		Torque - Max Backhand Distance (Nm)	Maximum Backhand Distance (m)
Torque - Max Backhand Distance (Nm)	Pearson		0.172
	Sig. (2-tailed)		0.331
	N		34
Maximum Backhand Distance (m)	Pearson	0.172	
	Sig. (2-tailed)	0.331	
	N	34	

Table 27

Pearson Correlation; Forehand Maximum Distance vs. Torque (RQ3)

Correlations			
		Torque - Max Forehand Distance (Nm)	Maximum Forehand Distance (m)
Torque - Max Forehand Distance (Nm)	Pearson		0.221
	Sig. (2-tailed)		0.208
	N		34
Maximum Forehand Distance (m)	Pearson	0.221	
	Sig. (2-tailed)	0.208	
	N	34	

Figure 13

Scatterplot; Backhand Maximum Distance vs. Torque (RQ3)

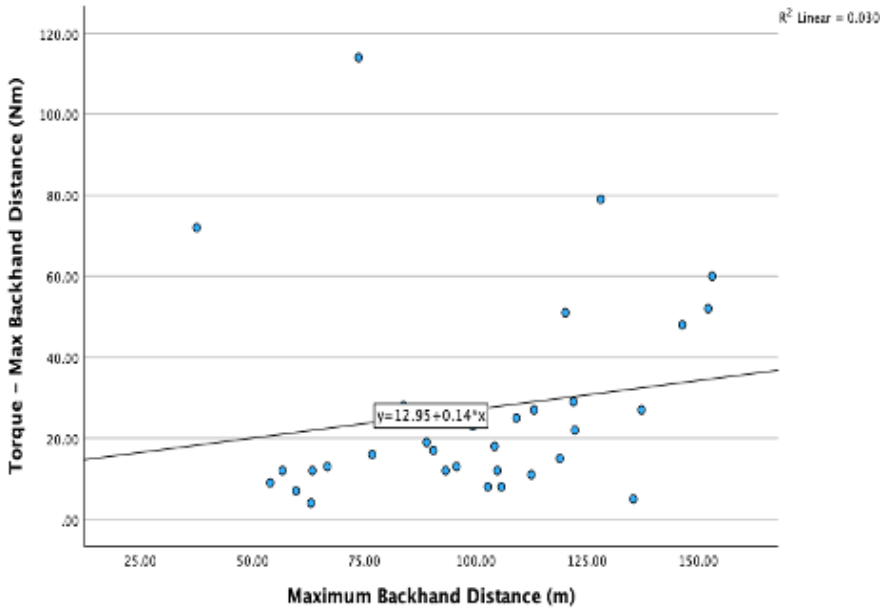
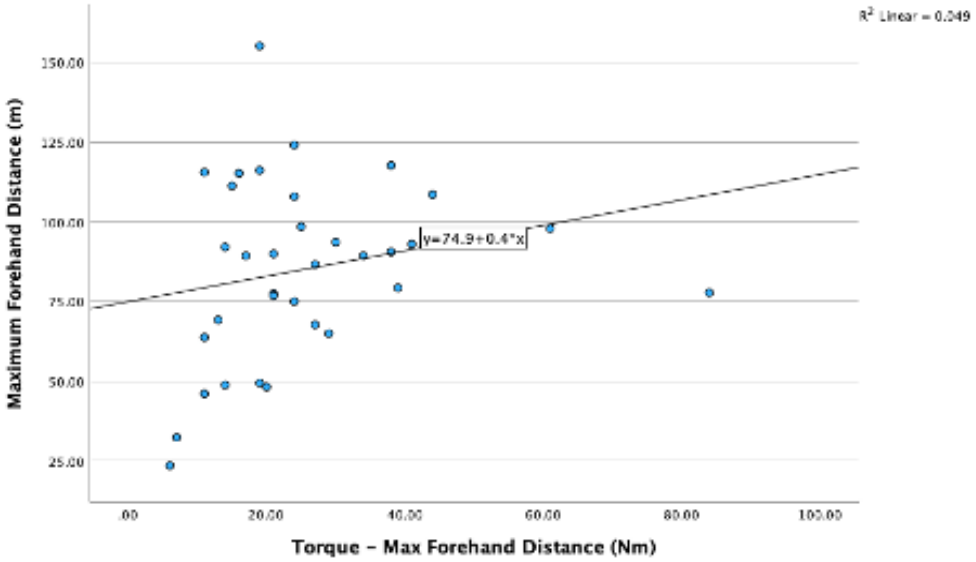


Figure 14

Scatterplot; Forehand Maximum Distance vs. Torque (RQ3)



APPENDIX B

LIBERTY UNIVERSITY.
INSTITUTIONAL REVIEW BOARD

July 31, 2024

Lauren Giles
Eric Sobolewski, Lauren Giles

Re: IRB Approval - IRB-FY23-24-2090 Quantification of Disc Golf Throwing Biomechanics and Relationship of Throwing Performance to Subject Demographics

Dear Lauren Giles, Eric Sobolewski, Lauren Giles,

We are pleased to inform you that your study has been approved by the Liberty University Institutional Review Board (IRB). This approval is extended to you for one year from the following date: July 31, 2024. If you need to make changes to the methodology as it pertains to human subjects, you must submit a modification to the IRB. Modifications can be completed through your Cayuse IRB account.

Your study falls under the expedited review category (45 CFR 46.110), which is applicable to specific, minimal risk studies and minor changes to approved studies for the following reason(s):

4. Collection of data through noninvasive procedures (not involving general anesthesia or sedation) routinely employed in clinical practice, excluding procedures involving x-rays or microwaves. Where medical devices are employed, they must be cleared/approved for marketing. (Studies intended to evaluate the safety and effectiveness of the medical device are not generally eligible for expedited review, including studies of cleared medical devices for new indications.)

For a PDF of your approval letter, click on your study number in the My Studies card on your Cayuse dashboard. Next, click the Submissions bar beside the Study Details bar on the Study Details page. Finally, click Initial under Submission Type and choose the Letters tab toward the bottom of the Submission Details page. Your stamped consent form(s) and final versions of your study documents can be found on the same page under the Attachments tab. Your stamped consent form(s) should be copied and used to gain the consent of your research participants. If you plan to provide your consent information electronically, the contents of the attached consent document(s) should be made available without alteration.

Thank you for your cooperation with the IRB, and we wish you well with your research project.

Sincerely,

G. Michele Baker, PhD, CIP
Administrative Chair
Research Ethics Office